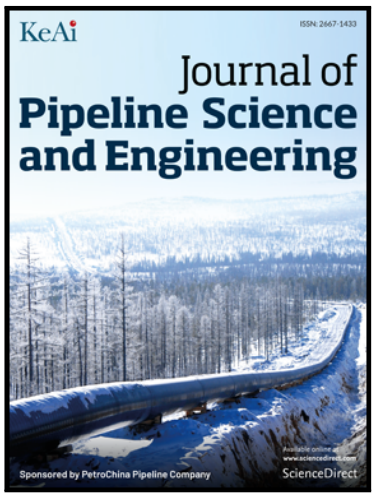
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State-of-the-art Machine Learning Advances in Reliability-based Design, Integrity Assessment, Inspection and Maintenance of Pipelines: A Systematic Review

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State-of-the-art Machine Learning Advances in Reliability-based Design, Integrity Assessment, Inspection and Maintenance of Pipelines: A Systematic Review

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A R T I C L E I N F O

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A B S T R A C T

Machine learning (ML) has increasingly been adopted in pipeline engineering, extending traditional physics-based, statistical, and empirical diagnostics toward data-driven frameworks. This review systematically examines ML applications in reliability-based design, structural integrity evaluation, condition monitoring, and inspection planning of pipelines. The literature shows a shift from conventional case-specific supervised learning toward transferable, hybrid, metaheuristic, and physics-informed techniques. These frameworks decompose signals, quantify uncertainty, use graph-based knowledge representation, and enhance generalizability through physics-informed learning, ranging from theory-guided features and architectures to limited forms of constraint enforcement. However, progress is limited by small, low-quality laboratory datasets, little focus on fatigue crack growth and multi-defect interactions, and heavy reliance on opaque surrogate models. A comparative analysis reveals persistent bottlenecks in generalization, interpretability, and the coupling of material, environmental, and operational variables. Three main research frontiers emerge: (1) large-scale, multi-source benchmark datasets, (2) physics-informed and interpretable learning frameworks that bridge domain mechanics and ML algorithms, and (3) standardized evaluation protocols and field-level validation schemes. The review concludes with a research roadmap to accelerate the move from algorithm-centric experimentation to tractable, trustworthy ML frameworks for pipeline integrity management. Overall, this study provides guidance for improving pipeline integrity to support safe and sustainable energy transport through resilient infrastructures.

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Nomenclature

Abbr.	Term	Abbr.	Term	Abbr.	Term
AE	Acoustic Emission	ELM	Extreme Learning Machine	LSTM	Long Short-Term Memory
AI	Artificial Intelligence	GAN	Generative Adversarial Network	MCS	Monte Carlo Simulation
ANN	Artificial Neural Network	GBM	Gradient Boosting Machine	MFL	Magnetic Flux Leakage
API	American Petroleum Institute	GBRT	Gradient Boosted Regression Trees	MFO	Moth-Flame Optimization

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BES	Bald Eagle Search	GAT	Graph Attention Network	ML	Machine learning
BiLSTM	Bidirectional Long Short-Term Memory	GNN	Graph Neural Network	SPH	Smoothed Particle Hydrodynamics
BNNs	Bayesian Neural Networks	HMOGWO	Hybrid Multi-Objective Grey Wolf Optimizer	SRF	Scalable Rain Forest
BPNN	Backpropagation Neural Network	FEA	Finite Element Analysis	SSA	Sparrow Search Algorithm
BRNN	Bidirectional Recurrent Neural Network	FEM	Finite Element Method	RF	Random Forest
CatBoost	Categorical Boosting	IGWO	Improved Grey Wolf Optimizer	RMSE	Root Mean Squared Error
CBA	Cost-Benefit Analysis	ILI	In-Line Inspection	SHAP	SHapley Additive exPlanations
CCTV	Closed-Circuit Television	IVIF	Interval-Valued Intuitionistic Fuzzy	STFT	Short-Time Fourier Transform
CDGM	Cross-Domain Generative Diagnosis Method	KNN	k-Nearest Neighbors	SWARA	Step Wise Weight Assessment Ratio Analysis
CWT	Continuous Wavelet Transform	LFS	Levy Flight Strategy	TGNN	Temporal Graph Neural Network
DDPM	Denoising Diffusion Probabilistic Models	LIME	Local Interpretable Model-Agnostic Explanations	TWSC	Strengthen Convolutional Neural Network
DNN	Deep Neural Network	LiDAR	Light Detection and Ranging	VMD	Variational Mode Decomposition
DNV	Det Norske Veritas	LSBES	Lévy Flight and Simulated Annealing Enhanced Bald Eagle Search		
DRL	Deep Reinforcement Learning	LSSVM	Least Squares Support Vector Machine		

1. Introduction

Pipelines form the backbone of urban infrastructure, conveying oil, gas, petrochemical products, water, and wastewater. Fossil fuels, particularly oil and natural gas, still dominate the global energy market, accounting for over 60% of primary energy consumption in 2024, with pipeline networks serving as the most economical and widely used infrastructure for their large-scale inland and near-shore transportation [1, 2]. Although proven reliable for long-distance fluid transport, pipelines face many damage mechanisms and external loads that challenge their useful life [3-5]. In addition to the internal pressure, onshore pipelines running above-ground are susceptible to blast [6] and third-party interference [4], whereas pipelines buried underground can suffer from impact loads [7], traffic loads [8], and geohazard-induced structural loads during both seismic and non-seismic events [9], combined with progressive general and localized corrosion due to the soil environment [10, 11]. Offshore pipelines are consistently exposed to seawater corrosion [12], anchor loads, and dynamic loads caused by waves or currents [13, 14], resulting in an increased rate of degradation. Rehabilitation, including section removal [15], composite sleeves [16-21], metallic repairs [22, 23], as well as proprietary underground retrofits [24] and replacement [25], can be effective when access is provided to mitigate the risk. However, failures cannot be ruled out largely because uncertainties still exist in the safety design, integrity assessment, and operation and maintenance [26-28]. In face of uncertainties, more sophisticated techniques using probabilistic approaches are used to form the basis for reliability-based design [18, 29, 30], pipeline integrity management (PIM) [31, 32], and risk-based inspection/maintenance planning [33-37].

Amidst continuing threats and the need of efficient uncertainty quantification, the past few decades have seen a surge in the application of artificial intelligence (AI), including machine learning (ML) as a subset of AI, to PIM. These technologies enable predictive maintenance, risk quantification, and autonomous inspection, with the potential for substantial improvements in safety, efficiency, and life-cycle costs. Methodological developments often outpace the actual implementation of methods in real-life situations, although recent review studies suggest that several challenges remain to be addressed. To synthesize recurring limitations reported in prior studies, this review identifies consensus gaps, reflecting

research challenges and methodological limitations consistently reported across multiple review articles. Based on a systematic screening of the literature, 24 relevant review papers were analyzed [37-60], from which ten recurring gaps were extracted and summarized by frequency in Table 1. Figure 1 shows the heatmap of consensus gap overlaps among 24 review articles to show their occurrence or co-occurrence, highlighting the key shortcomings that tie these gaps together across utilities as diverse as oil and gas transmission [38-50], water distribution networks [51, 52], and drainage infrastructure [53, 54].

The most identified gap is the lack of proper datasets and quality issues of benchmarks [39, 41-43, 45, 47, 51, 52, 54-56]. Reproducibility and fair model comparison will not be achieved as long as data formats, features, or performance metrics remain scattered; this is a concern already observed, e.g., in the modeling of corrosion [39], structural health monitoring [42, 57], and two-phase flow prediction in drainage [54]. This limitation is exacerbated by another widespread gap highlighted in many reports, namely the lack of validation in real-life environments [39, 41-43, 45, 47, 51, 52, 54-56]. The literature is dominated by laboratory-bound prototypes and virtualized models, with limited studies dealing with the operational complexity resulting from live networks (e.g., environmental constraints for unmanned aerial vehicle inspection schemes [38], regulatory aspects for offshore asset integrity management [41], stress corrosion cracking prediction [43], corrosion growth models [46]). Validation becomes more challenging due to the lack of universal evaluation protocols, benchmark datasets, and testing frameworks, which makes it difficult to fairly compare the performance of ML techniques [39, 41-43, 45, 47, 51, 52, 54-56].

A further cross-cutting obstacle is the black-box nature of most ML architectures [39, 41, 42, 45, 47, 51, 54, 55, 58]. Reasoning opacity in the developed model challenges regulatory acceptance and operator trust, especially for decisions regarding critical safety-related issues. Therefore, interpretable approaches such as physics-informed networks, feature attribution, and hybrid mechanistic techniques are being increasingly developed (e.g., Industry 4.0 structural health monitoring [58]). In addition, despite the increasing availability of sophisticated hardware, multi-sensor integration remains underutilized [38, 40-42, 51, 53, 59, 60]. The majority of deployments emphasize a single modality, comparable to the use of optical unmanned aerial vehicle images [38] or single-mode structural health monitoring [42], rather than merging, for example, visual, acoustic, and fiber-optic strategies into integrated analytics pipelines. The lack of multi-sensor integration is reflected in the continued existence of bottlenecks to effective performance due to computational scalability constraints [39-42, 45, 54, 55, 60], where deep models still fail to meet even real-time needs, and they cannot handle network-scale datasets with the available resources.

Narrow subsystem focus has been a matter of concern in several studies [38, 40, 44, 51, 53, 58, 61]; even highly evolved solutions are typically cast independent of the broader lifecycle contexts and thus fail to reach the design-through-decommissioning perspectives required for comprehensive integrity management. A closely related gap is cross-domain generalization [39, 41, 43, 45, 51, 52], which also restricts scalability, thereby limiting its reliability. For example, a corrosion model optimized in one geography or with a specific material performs unreliably when applied to other environmental conditions or different material compositions. Uncertainty quantification is also underrepresented [41, 42, 47, 58, 62], although it is a key issue in risk-aware asset management. Outside of the technical space, there are a few studies/insights on regulations, ethics, and operationalization paths to adoption [41, 45, 51, 58, 61], with open questions regarding safety certification processes, governance mechanisms, and industrial uptake.

In summary, the ten frequently identified gaps and their patterns of co-occurrence provide an evidence-based overview of recurring limitations reported across the reviewed studies and help inform priorities for interdisciplinary research. As illustrated in Figure 1, several gaps are frequently discussed together in the literature, which indicates common patterns. For example, dataset scarcity and limited model interpretability are often jointly noted, reflecting practical trade-offs encountered in engineering applications where constrained data availability can motivate the use of more complex, less transparent models. Similarly, the repeated joint discussion of missing standards and limited field validation highlights persistent barriers to deployment. Addressing these co-occurring but conceptually distinct limitations will be essential for advancing ML-based PIM technologies from laboratory-scale demonstrations toward reliable industrial implementation.

Most ML-related review papers focused primarily on one lifecycle stage in isolation. To address the fragmented treatment of ML applications in the literature, this study adopts a lifecycle-based framework. Herein, the pipeline lifecycle is defined as comprising three stages: (1) reliability-based design and safety assessment [38, 44, 56], (2) structural integrity and degradation modeling [39, 45, 46, 48, 50, 55, 59, 62], and (3) pipeline inspection and maintenance planning [40-42, 45, 47, 49, 51-53, 58, 60, 61]. Stage 1 involves quantifying failure probabilities and establishing safety margins under modeling and data uncertainty, while stage 2 focuses on predicting the progression of defects such as corrosion, cracking, and mechanical damage, often under nonlinear degradation mechanisms and heterogeneous data sources. Finally, stage 3 deals with detecting existing defects, assessing current conditions, and scheduling cost-effective interventions, where challenges related to data quality, interpretability, and practical deployment are particularly pronounced.

Table 1. Consensus gaps in the review articles: frequency and representative articles (n = 24 reviews).

Gap cluster	Description	Frequency n (%)	Representative articles
Dataset scarcity and quality issues	Incomplete, biased, or noisy datasets; class imbalance; restricted access to field measurements; limited temporal/spatial coverage.	13 (54%)	[39, 41-43, 45, 47-51, 54, 55]
Limited real-world validation	Predominantly lab-based or simulation studies; minimal deployment in operational environments.	12 (50%)	[38, 41-44, 46, 47, 49, 50, 56, 59, 62]
Lack of standardized evaluation protocols	Absence of unified performance metrics, benchmark datasets, or testing frameworks; prevents reproducibility and fair comparison.	11 (46%)	[39, 41-43, 45, 47, 51, 52, 54-56]
Model interpretability deficits	Over-reliance on black-box ML models without clear feature attribution or physical linkage.	9 (38%)	[39, 41, 42, 45, 47, 51, 54, 55, 58]
Limited multi-sensor integration	Sparse fusion of heterogeneous sensor streams (e.g., optical, acoustic, fiber-optic) into unified analytics.	8 (33%)	[38, 40-42, 51, 53, 59, 60]
Computational scalability constraints	Algorithms too slow or resource-intensive for real-time or network-scale application.	7 (29%)	[39-42, 45, 54, 55, 60]
Narrow subsystem focus	Solutions address single components without integrating full asset lifecycle or system interactions.	7 (29%)	[38, 40, 44, 51, 53, 58, 61]
Weak cross-domain generalization	Model performance degrades when transferred to different regions, materials, or asset types.	6 (25%)	[39, 41, 43, 45, 51, 52]
Lack of uncertainty quantification	No statistical confidence bounds on predictions; omission of risk-aware decision metrics.	5 (21%)	[41, 42, 47, 58, 62]
Neglect of regulatory/operational adoption pathways	Minimal treatment of safety standards, ethics, certification, operator training, or scaling.	5 (21%)	[41, 45, 51, 58, 61]

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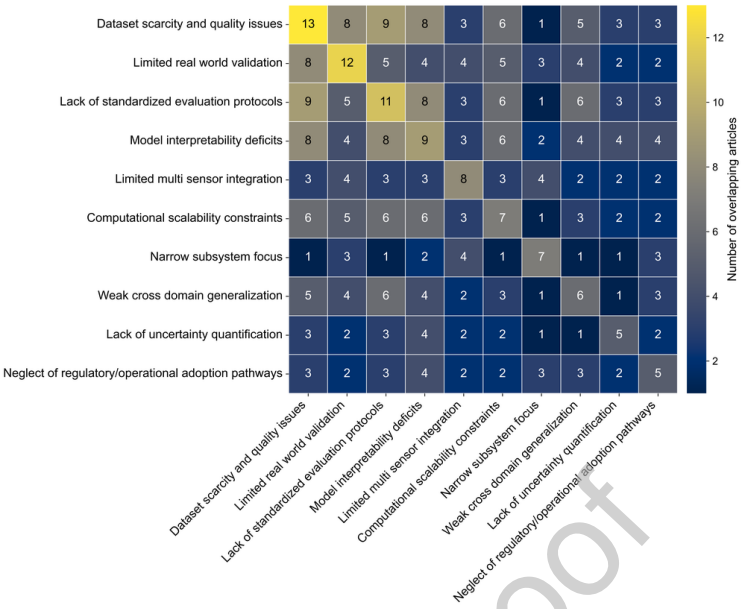


Figure 1. Heatmap of consensus gap overlaps among 24 review articles.

The present review systematically organizes ML-based original studies according to the above lifecycle stages. This structure enables the comparison of learning objectives, data requirements, dominant algorithms, and reported challenges at each stage, while also revealing cross-cutting limitations such as data scarcity, lack of standardization, limited interpretability, and insufficient field validation. By establishing this lifecycle-oriented synthesis, the review provides a coherent framework for benchmarking methodological progress and identifying research priorities that support end-to-end deployment of ML-based PIM solutions. As such, this study identifies the most and least effective algorithms for context-driven applications. These findings are reinforced by a comparative study, which also provides directions for further research. The unified analysis is designed to promote both methodical rigor and practical deployment, providing a common platform for engineers working in ML-driven pipeline integrity management.

2. Methodology

The methodological framework for this systematic review is organized as follows. Initially, the scope and principal concepts of ML in pipeline engineering are defined, with attention to relevant lifecycle stages and associated engineering challenges, followed by a detailed description of the literature search strategy and selection criteria.

2.1. *Preliminaries and scope definition*

In this review, ML refers to data-driven computational methods that infer functional relationships from engineering datasets to enhance pipeline reliability, integrity assessment, and maintenance decision-making. The scope includes classical supervised learning techniques, including artificial neural networks (ANNs), support vector machines (SVMs), and decision trees, as well as ensemble and tree-based models, deep learning (DL) architectures, Bayesian and uncertainty-aware models, and hybrid multi-stage frameworks that incorporate optimization, signal decomposition, or physics-based constraints. Physics-informed, surrogate, and transfer-learning approaches are considered when learning-based components are central. Conversely, purely rule-based expert systems and deterministic numerical simulations without adaptive learning are excluded because they represent established baseline practices in PIM. Their role is

nevertheless reflected through the review of hybrid and ML-augmented frameworks, where legacy rules and models are enhanced using data-driven and uncertainty-aware methods.

This review includes both onshore and offshore pipeline systems utilized for transporting oil, gas, water, or multiphase fluids. The analysis is limited to engineering challenges that directly impact PIM throughout the asset lifecycle, including reliability-based design, structural integrity and degradation assessment, and inspection and maintenance planning. The pipeline lifecycle is divided into three stages:

1) Reliability-based design and safety assessment: This stage focuses on uncertainty quantification, failure probability estimation, and risk-informed decision-making, with the objective of supporting safe and reliable pipeline design under variable material, loading, and environmental conditions.

(2) Structural integrity and degradation modeling: This stage addresses the modeling and prediction of damage mechanisms, including corrosion, cracking, mechanical damage, geohazards, and residual strength degradation, to assess pipeline performance throughout its operational life.

(3) Pipeline inspection and maintenance planning: This stage involves defect detection, condition monitoring, inspection optimization, and maintenance decision support, aimed at enabling cost-effective integrity management and timely intervention strategies.

2.2. *Literature search and selection process*

This study adhered to the PRISMA guidelines [63] for the selection of references. Databases, processed article counts, and selection criteria are detailed below:

(1) Literature databases for search: Scopus and Web of Science were used as the primary sources due to their broad coverage of engineering publications and established citation indexing. Searches were limited to English-language journal articles published between 2015 and 2026 under engineering subject areas.

(2) Keyword selection: Three thematic groups of keywords were defined to correspond with the structure of this study. The first group was within the scope of pipeline design and safety evaluation, combining ML terms such as “machine learning”, “artificial intelligence”, and “deep learning” with probabilistic and reliability descriptors including “probabilistic”, “uncertainty”, “risk assessment”, and “safety”. The second group was within the scope of pipeline integrity assessment, combining ML terms with surrogate modelling and physics-based analysis descriptors such as “surrogate model”, “response surface”, “finite element”, “meta-model”, “integrity”, and “geohazard”. The third group addressed pipeline inspection and maintenance planning, joining ML terms with maintenance and geospatial descriptors such as “maintenance”, “inspection”, “landslide”, “GIS”, and “LiDAR”. By joining ML terms with those from each thematic range, targeted search strings were developed to maximise relevant literature retrieval.

(3) Inclusion and exclusion criteria: Articles had to employ ML approaches that directly addressed pipeline reliability, integrity, or inspection/maintenance planning, with explicit computational modeling or predictive analysis based on engineering datasets. Studies from unrelated domains or those lacking a quantitative evaluation of model performance were excluded.

(4) Screening process: After database searches, records were imported into EndNote [64] for de-duplication and applying search filters. The pipeline design and safety evaluation search resulted in 78 records, which were reduced to 29 after abstract screening. The pipeline integrity assessment returned 54 post-filtered records, with 42 retained after reading abstracts. The pipeline inspection and maintenance planning contained 40 post-filtered records, which were narrowed to 21 following a detailed abstract review. Screening evaluated titles, abstracts, and keywords for thematic fit; full-text reading was performed to determine final inclusion. Key bibliographic and methodological information (publication year, authors, algorithms, performance metrics, input data type, and reported limitations and advantages) was recorded for analysis. Of the 3042 initial sources identified prior to de-duplication and selection, 57% came from Web of Science, 42% from Scopus, and 1% from a hand search. After rigorous filtering and screening, 95 articles were retained.

Notably, 90% of the studies ultimately included in the review were identified using database searches, while the remaining 10% were identified through targeted hand searches.

(5) Quality assessment: Only peer-reviewed journal articles were included, with preference for those published in journals indexed in Web of Science or Scopus, or recognised as leading publications in the pipeline engineering and reliability domain.

Data from the selected 95 sources were imported into VOSviewer for keyword analysis, detailed in Section 3, and then systematically reviewed in Sections 4-6.

3. Scientometric analysis

An aggregation of keyword co-occurrence patterns in the 95 studies was outlined to create a networked thematic landscape. Top links, ML, and risk assessment are, by far, the most popular linkages involving ML, highlighting risk assessment as a key concern for predictive maintenance and decision-support systems. Corrosion and underground pipe deterioration were closely related clusters that represented ongoing interest in pipeline integrity. Another cluster, focusing on pipeline integrity, performance metrics, and modeling approaches (e.g., FEA), represented a cornerstone for the engineering basis of this research field. The density gradients revealed the recent intertwining of computational intelligence in corrosion prediction as an emerging trend, while weaker peripheral surrounds, such as offshore pipeline behaviour, reflected less explored niches. The level of thematic overlap suggested that notable progress could be expected from hybrids utilizing well-established engineering methods in conjunction with adaptive, data-driven analytics.

Figure 2-Figure 4 visually plot the thematic emphasis, temporal dissemination pattern, and conceptual intersections in ML techniques employed in pipeline research. The keyword occurrence heatmap (Figure 2) clearly shows that terms related to the assessment of pipeline integrity, particularly ML risk assessment and corrosion, were the most popular among discourse elements, indicating that predictive diagnostics has been given a higher priority than early-stage design or post-installation inspection. The temporal evolution of topics in Figure 3 demonstrates a sharp increase in data-driven and computational issues after 2022, with ML and related terms becoming predominant quickly, implying an emerging technological shift towards algorithmic robustness evaluation. Typical concepts such as corrosion are part

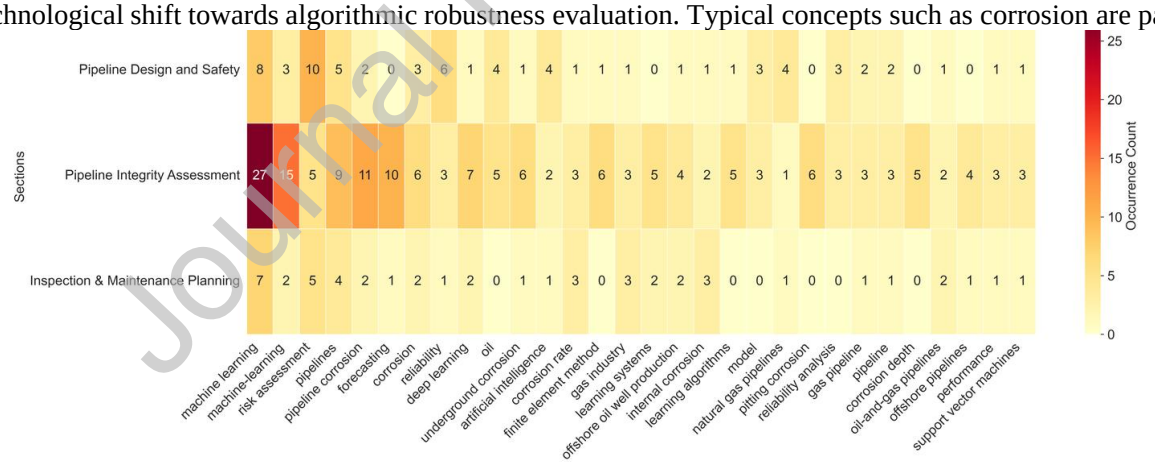


Figure 2. Keyword occurrence heatmap for top 30 VOSviewer keywords.

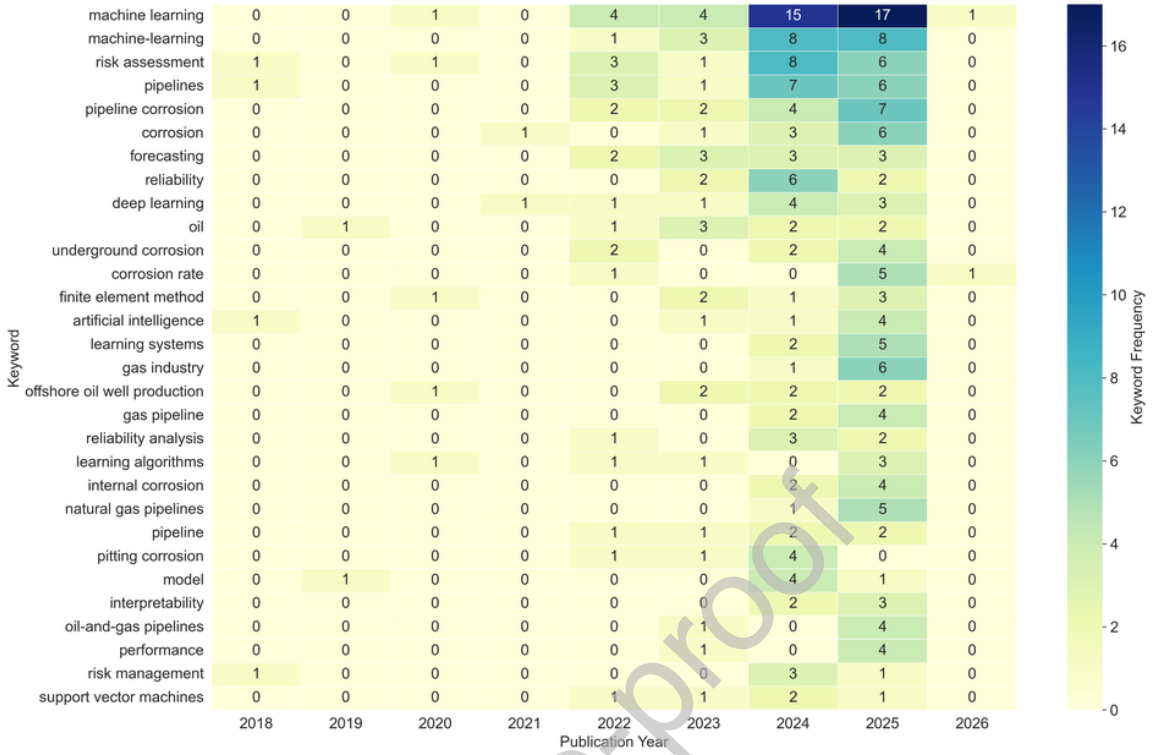


Figure 3. Temporal keyword evolution for top 30 VOSviewer keywords.

of logistics, but with an increasingly greater connection to digital methods. The overlap analysis in Figure 4 indicates that the domain related to pipeline integrity assessment is the most interrelated, with significant keyword links to both design and inspection stages, serving as a primary interface for translating design advances into operational safety and informing inspection strategy decisions. Such an integrated perspective implies that future research impact will rely on intensified cross-stage continuity, particularly by embedding advanced analytics throughout the full pipeline lifecycle.

The color-coded word cloud (Figure 5) maps the interlinked thematic priorities that emerge across the three stages of pipeline studies. Pipeline design and safety evaluation (red) overlaps with pipeline integrity assessment regarding reliability assessment, while there are some unique keywords, such as fuzzy logic, and pipeline safety, reflecting quantitative risk frameworks and stochastic modeling for design

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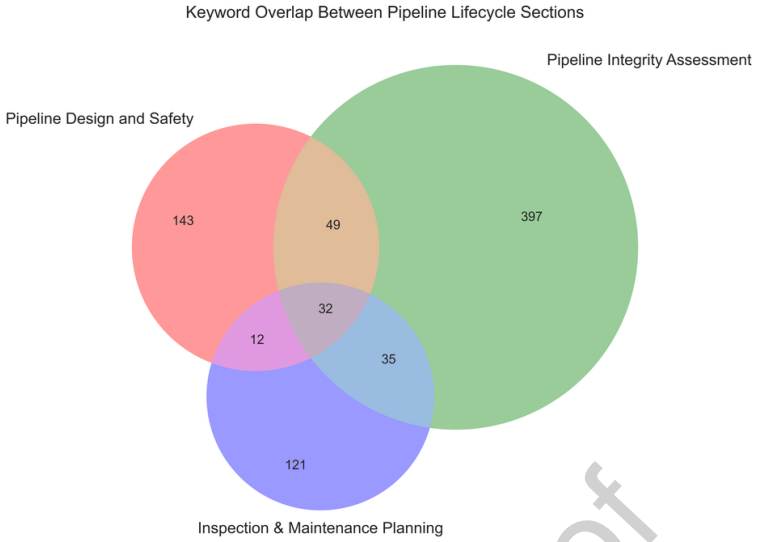


Figure 4.Overlap of keywords between studies in three different stages of design, assessment, and inspection.

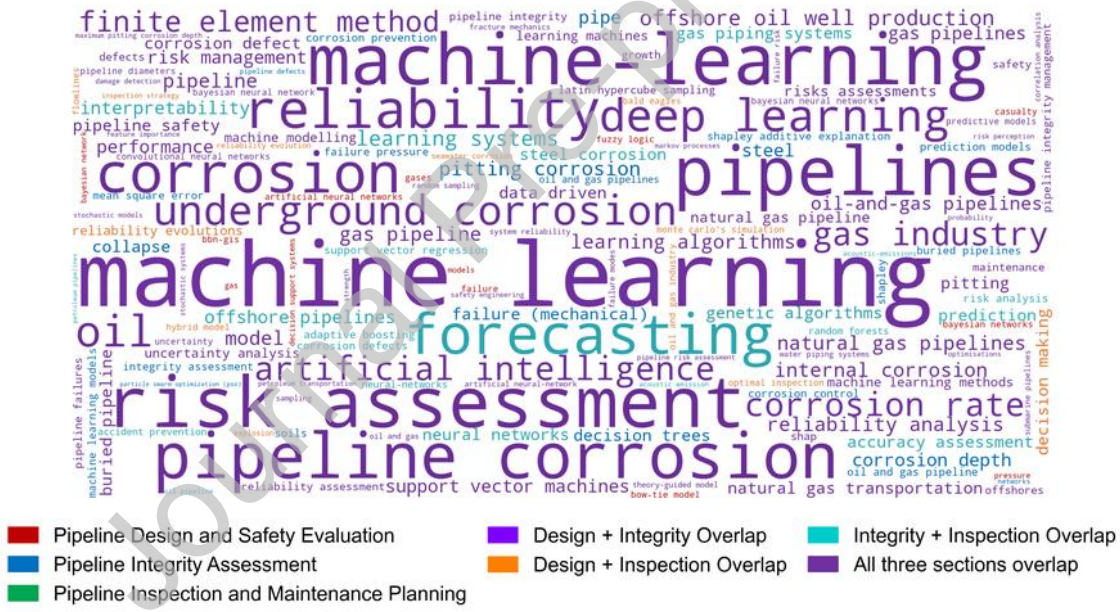


Figure 5.Word cloud for keywords of original studies.

validation. Pipeline integrity assessment (blue) focuses on degradation mechanisms and predictive surrogates, with high-frequency terms such as corrosion, offshore pipelines, and burst pressure prediction. This shows the emphasis on the assessment of corroded assets through numerical simulation, defect characterization, and data-driven sampling for ML calibration. Pipeline inspection and maintenance planning (green) highlights operational readiness and asset life-cycle control, with forecasting, risk management, corrosion growth rate, and decision-making demonstrating a shift toward predictive analytics and AI-assisted scheduling. FEM has been shared between pipeline design and integrity assessment, demonstrating its application in generating samples at these two stages, with new opportunities for integrating NNs into such techniques for generalized, interpretable surrogate modeling. Quantified overlaps

common between all three sections, including ML, AI, and SVMs, highlight the integration of intelligent modeling across the pipeline lifespan, design optimization, integrity evaluation, and maintenance planning. This synthesis provides a thematic framework for the methodological assessments presented in the subsequent sections, where cross-sectional keyword patterns will be examined within their applied contexts.

4. Pipeline design and safety evaluation

Pipeline design and safety assessment studies are converging on integrated approaches that incorporate quantitative reliability assessment, probabilistic risk analysis, and state-of-the-art computational modeling. Four key themes (reliability, risk, corrosion, and failure) emphasize the critical impact of degradation mechanisms upon the safety-critical performance. The widespread integration of intensive data management techniques, such as NNs, Bayesian inference, and fuzzy logic, with fundamental engineering principles showing a transition from deterministic methodologies to hybrid, resilience-based concepts. This evolution encompasses a range of media and operating conditions, from gas and hydrogen to offshore systems, requiring a design concept capable of withstanding multi-hazard loads while also satisfying regulatory and sustainability constraints. The following subsections discuss recent developments in the application ML techniques in reliability and probabilistic safety methodologies, risk-based design strategies, and service-life prediction models.

4.1. *Reliability and probabilistic safety frameworks*

The integration of ML algorithms with existing reliability and probabilistic techniques has led to progress across a wide range of pipeline-related safety topics (Table 2) [65-72]. By replacing, enhancing, or complementing classical probabilistic methods with more nuanced data-driven models, such studies have demonstrated improvement in predictive accuracy, computational efficiency, and measurable insight, as well as actionable value, which cannot be achieved using purely analytical approaches.

In seismic reliability evolution [65], the surrogate model combining Levy flight strategy (LFS), sparrow search algorithm (SSA), and Backpropagation NN (BPNN) (LFS-SSA-BPNN) integrated FE simulations with a metaheuristically optimized NN, achieving R^2=0.952 and improving accuracy by 0.089 compared to the SSA-BPNN benchmark, while reducing error variance. Maintaining the accuracy of Monte Carlo Simulation (MCS), this technique eliminated the prohibitive runtime of repeated limit state function evaluations. Similarly, in the reliability assessment of corroded pipelines [66], the use of Extreme Learning Machine (ELM) optimized via the Levy Flight and Simulated Annealing Enhanced Bald Eagle Search (LSBES) algorithm resulted in an advanced heuristic optimization framework which delivered R^2=0.9926 and mAP=94.83%, a 6% gain over BES-ELM, enabling long-term reliability evolution analysis from field data without sacrificing stability. Extending this progression toward spatial reliability mapping [67], the Runge-Kutta4-Net framework demonstrated near-perfect agreement ($R^2 = 0.9998$) while achieving approximately 200× computational acceleration compared to MCS, yet remained constrained by idealized defect geometries and lack of interaction modelling; an area where multi-defect coupling and physics-guided learning can enhance generalizability.

In cases where there is a lack of sufficient data, synthetic data can be generated by integrating ML techniques into probabilistic modelling. In burst-failure risk analysis [68], a Gaussian Copula Generative Adversarial Network (GAN) generated synthetic, physically consistent corrosion defect data, improving the accuracy of the random forest (RF) model from 86% to 96%. For corrosion failure risk under varied steel grades [69], coupling ANN with MCS and SHapley Additive exPlanations (SHAP) interpretability achieved 93.5% accuracy, training in only 29.27 seconds. It provided justifiable, interpretable risk drivers, offering both operational speed and transparency for decision-makers.

In qualitative or semi-quantitative domains, integrating ML enhances traditional risk models. In oil pipeline risk scoring [70], reliability logic was embedded in a fuzzy inference system, which enabled the representation of uncertainty from expert judgment while translating outputs into explicit cost-risk ranges

Table 2. Comparative performance of ML-based probabilistic frameworks.

Reference	Application focus	Method / Algorithm	Data (type, size)	Key performance (proposed vs. benchmark)	Major advantages	Notable novelty	Limitations
[65]	Seismic reliability evolution	LFS-SSA-BPNN surrogate + FE + MCS	FE simulations: 180 scenarios	LFS-SSA-BPNN: R2 0.952, RMSE 2.96, MAPE 5.65%, Acc 94.7% SSA-BPNN: R^2 0.863 BPNN: R^2 0.378	+ Accuracy ↑ by +0.089 R^2 over SSA-BPNN + High stability; narrow RE range	LFS integration into SSA for ANN weight optimization	Single seismic wave; fixed failure criterion
[66]	Corrosion-driven reliability evolution	LSBES-ELM + MCS	Field corrosion: 246 samples	LSBES-ELM: R^2 0.9926, mAP 94.83%, RMSE 0.576 BES-ELM: R^2 0.9307 ELM: R^2 0.8170	+ mAP ↑ 6% over BES-ELM + Stable, smallest RE range	Enhanced BES w/ Lévy flight + SA	Sensitive to noisy and incomplete datasets; fixed failure criterion
[67]	Reliability mapping for defects	MFL + Runge-Kutta 4-Net; FE-derived labels	FE; 3750 samples; cuboid/ellipsoidal defects	R^2 0.9998 (RK4-Net) vs. 0.9997 (Euler-Net)	~200× faster than MCS	Direct MFL→reliability mapping	Simulated shapes; ignores defects interaction
[68]	Burst failure risk, data augmentation	GC-GAN + RF	Exp. data: 92; GAN = 100 synthetic pts	Mixed data model: Acc 96%, AUC 0.99 Real-only: Acc 86%, AUC 0.96	+ GAN ↑ accuracy by +10% + Solves data scarcity	GAN-based synthetic pipeline failure dataset	Needs initial real data; neglected geological hazards and third-party events
[69]	Corrosion failure risk under varied grades	ANN + MCS + SHAP	Sim: 6000; Validation: 80 PHMSA	ANN: Acc 93.5%, Train 29.27s SVM: 85.42% XGB: 84.25%	+ Fast (<0.01 s prediction) + Interpretable via SHAP	SHAP-interpreted hybrid ML-physics risk model	Single defect assumption; param correlation ignored
[70]	Oil pipeline risk score	Fuzzy Inference System (Mamdani)	Field expert data; 16 km, 8 segments	Mean error ↓ vs. traditional; cost risk $1.27k-$1.27k-$2.15M	+ Handles uncertainty; captures expert knowledge	Monetary risk quantification in fuzzy risk model	Subjective rule weighting; limited validation data
[71]	Urban gas leakage probability	Interpretive Structural Modeling-Bayesian Network + EM	Accident reports: 89 cases	Verification: predicted leak prob 64% (high-risk, matched actual)	+ Sensitivity shows targeted prevention reduces risk 40-48%	Interpretive Structural Modeling to inform BN structure on real accident data	Small dataset; coarse state variables

(USD 1.27k-2.15M), moving beyond static categorical assessments. Likewise, regarding the leakage probability of urban gas pipelines [71], an Interpretive Structural Modeling-informed Bayesian Network captured causal dependencies from accident data. This technique produced scenario-specific leak risk

predictions and quantified the impact of targeted mitigations (up to 48% reduction from human-error interventions).

Together, these cases provide three main insights. First, ML-enhanced reliability frameworks maintain probabilistic soundness while dramatically reducing computational costs; that is, numerous parameter combinations can be evaluated. Second, optimization heuristics and generative models address several critical operational issues, enhancing stability in the presence of noisy or scarce data, and broadening applicability to rarely observed types of hazards. Third, by combining interpretability tools with consequence quantification, computational models are linked with pragmatic safety strategies, rendering these methods technically sound and operationally compelling.

Throughout the reviewed studies [65, 66, 68-71], most of the gap clusters are highly correlated with several prior reports, such as dataset scarcity and quality limitations [65, 66, 68, 71], corrosion field datasets [66], small accident datasets [71], synthetic generation [67, 68], lack of standardized evaluation protocols [65, 66, 69], limited real-world validation [65, 69, 71], and persistent model interpretability challenges, addressed only by Xiao et al. [69] through SHAP explainability. The computational scalability is also partially limited, despite gains from surrogate modeling [65, 66, 69], as opposed to weak cross-domain generalization and narrow subsystem focus, which are evident in all references [65, 66, 68-71]. There are some less directly addressed issues, including multi-sensor integration, explicit model uncertainty quantification, and pathways for regulatory or operational adoption, only available to a limited extent [70, 71]. Finally, a new gap cluster is emerging: hazard-specific overfitting in hybrid ML-reliability frameworks, where optimization targets are narrowly tuned to application hazards (e.g., single seismic wave [65], single-defect assumption [67-69]), preventing transfer across multi-hazard and/or multi-defect conditions. Acknowledgement and mitigation of this limitation is necessary to develop generalizable, field-applicable predictive frameworks.

4.2. *Risk-based techniques*

The discussed studies reveal a wide range of methodologies for risk-based decision-making in the safety assessment of pipeline systems, with specific trade-offs in terms of accuracy, applicability, and data input [73-80] (Table 3). The multi-criteria approaches [73, 74] can be effective in limited data scenarios, particularly when expert judgement and hybrid decision frameworks are employed. Toufighi et al. [73] employed intuitionistic fuzzy computing, utilizing Interval-Valued Intuitionistic Fuzzy Step-Wise Weight Assessment Ratio Analysis (IVIF-SWARA) to assess dynamic risk factors. Hassan et al. [74] utilized an evidential reasoning procedure to merge and optimize cost-benefit analysis (CBA), thereby facilitating communication between intervention options and computable economic gains. Both studies [73, 74] excelled at incorporating uncertainty and trade-offs; however, they inherently relied on subjective expert judgment for prioritizing accuracy [73], and for parameterizing costs and benefits [74], which constrained reproducibility across environments.

However, data-rich ML-driven models obtained impressive predictive performance [75-77] (e.g., Ouadah [77] for large-scale In-Line Inspection (ILI) datasets; Vanitha et al. [75] achieving 97.5 % accuracy with fast inference). Squeeze-and-Excitation ResNet-50 extensions for dust-explosion risk in dedusting pipelines achieved similar accuracy [81], while differing in scope and generalizability. Vanitha et al. [75] developed a Rough Set Theory-based ML technique for large-scale qualitative risk scoring, while Campari et al. [76] integrated Gradient Boosting into the American Petroleum Institute (API) 580/581 Risk-Based Inspection framework, a development particularly important for hydrogen-related fatigue, while limited by class imbalance. The hybrid methodology proposed by Ouadah [77] combining unsupervised and supervised techniques excelled in prioritizing metal-loss defects but overlooked other defect morphologies and external benchmarks.

The probabilistic and dynamic system models [78, 79] expanded the scope of design spaces by integrating Bayesian reasoning into operational decision-making. The DRL-Bayesian network developed by Fan et al. [78] dynamically optimized reliability in gas pipeline networks, outperforming predefined and

Table 3. Comparison of ML techniques integrated in pipeline risk assessment.

Reference	Application focus	Method / Algorithm	Data (type, size)	Key performance (proposed vs. benchmark)	Major advantages	Notable novelty	Limitations
[73]	Risk-based maintenance design for oil transfer pipelines	Intuitionistic Fuzzy Computational Approach + IVIF-SWARA + FMEA + HFPR + IVIFPHA	Expert panel opinions (n=20) + OREDA data; 25 equipment items	Cost: considerable reduction vs. isolated strategies; Risk weight range: 0.412 to -0.019	Handles uncertainty, optimal maintenance prioritization, reduces bias	Integrates Fuzzy Computational Approach with IVIF-SWARA for dynamic, uncertain pipeline risk management	Subjective expert reliance; difficulty in accurate assessments
[74]	Decision support for Risk Control Options in data-scarce environments	Evidential Reasoning (ER) + Cost-Benefit Analysis (CBA) + AHP	Expert elicitation + literature + 10-year failure stats; Nigeria cross-country pipeline	Annual failure cost: $712.6M (baseline); Net benefit: $560.67M (7 best Risk Control Options); Likelihood reduction: 0.3000→0.0770	Handles uncertainty in hybrid data; clear operator guidance; adaptable framework	Combines ER with CBA tailored for intentional damage in low-data contexts	Small expert pool; assumes CBA parameters; focuses on likelihood not consequences
[75]	Qualitative risk assessment via Relative Risk Score	Relative Risk Scoring+ Naive Bayes + Decision Tree + SVM + Graph Convolutional Network	Kaggle dataset: 54	Accuracy: 97.5%vs. SVM 92%, Naive Bayes 92.7%, GCN 93%; Precision: 0.95; Recall: 0.93; Runtime: 24 ms	High accuracy, fast execution, broad parameter coverage	Developed Relative Risk Score ML model assigning numerical risk scores for comprehensive assessment	Requires further experiments
[76]	Risk-based inspection for hydrogen-enhanced fatigue crack growth	GBM Classifier within API 580/581 Risk-Based Inspection framework	Lab fatigue crack growth rate test data from Sandia National Labs; size NR	Accuracy: 94.61%(overall); Precision: 71.43-96.77%; Recall: 66.67-99.16%; AUC PR: 82.0-99.68%	Superior prediction over tree-based algorithms, robust to outliers, adaptable database	First integration of GBM into Risk-Based Inspection for hydrogen fatigue not in standards	Class imbalance lowers extreme/high predictions; computationally intensive
[77]	Defect risk assessment and prioritization	K-means + Decision Trees + NNs + AHP	ILI data (~40000 metal-loss defects)	Decision Tree: 99.80%train / 99.27%CV; NN:	Combines unsupervised/supervised models with MCDM for prioritization	Hybrid K-means + Decision Tree/NN + AHP for	Limited to metal loss defects; excludes other pig data; no

		prioritization		98.85%train / 98.35%CV; AHP CR: 0.06		defect risk ranking	external comparisons
[78]	Gas supply reliability optimization in natural gas pipeline networks	Bayesian Networks + Deep Reinforcement Learning (DRL) + MDP + Graph Theory + MCS	European NGPN case; 692.8 km; 2 sources, 2 compressor stations, 8 consumers; simulated 48 months	System reliability: DRL 0.9603 vs. scheduled 0.9167 & knowledge-based 0.849; Variance: DRL 0.07 vs. 0.13; Training: ~4.2h	Dynamic adaptation via online learning; system-level optimization	Integrates BN with DRL to jointly optimize reliability and maintenance cost	Simplified cost model; assumes perfect maintenance; static demand
[79]	Risk control for urban gas pipeline leaks/explosions	Bayesian Networks + Influence Diagrams + Event Sequence Diagrams + Fuzzy-set Theory	Case study (Shiyan gas explosion, China) + historical/environmental data	Strategy 12: Cost ¥1.76×10^4; Fire prob ↓ 23.1%, Explosion ↓ 23.0%; Strategy 14: Cost ¥4.06×10^4; Fire prob ↓ 24.6%	Captures interdependencies, integrates failure and consequence models, supports cost-effective solutions	Unifies BN failure model with accident consequence evolution for decision-making	Single-case generalizability; subjective expert inputs; assumes BN causal completeness

knowledge-based benchmarks, but with a high computational cost and simplified demand/cost assumptions. Li et al. [79] applied Bayesian networks integrated with influence diagrams to link failure causes to consequence progression and target leak and explosion suppression measures. Both studies included reciprocal relationships, although the single-case focus is one of the limitations reducing the transferability of their findings, as opposed to higher adaptability through simulated exploration of system states achieved by the former study [78].

Systemic challenges arise due to a lack of available datasets and inconsistent evaluation protocols, which can hinder the development of high-accuracy models [75, 76]. Most studies are based on simulations, laboratory data sets, or single-case examples, which hampers external generalization. Model interpretability drawbacks, especially relevant for hybrid DRL-based techniques [78] or multi-stage fuzzy logic-based approaches, further hinder the adoption process in operational settings. Scalability for field-level applications is infeasible due to computational limitations, particularly for resource-heavy models such as Gradient Boosting Machine (GBM) [76] and DRL [78]. While uncertainty quantification is handled in Bayesian-inspired frameworks [78, 79], the majority of ML-centric approaches circumvented explicit probabilistic treatment, thereby compromising the risk communication. Moreover, CBA was not adequately addressed or considered, except in Ref. [74], and class imbalance is an overlooked limiting factor in defect prediction [76]. Hybridization strategies are the most promising way forward: incorporating high-accuracy ML predictors into probabilistic, economically aware frameworks that can also gracefully push to the boundaries of incomplete or conflicting knowledge, providing transparent, operationally practical decision support.

4.3. *Service life prediction and degradation modelling*

Most studies focused on estimating the life of corroded pipelines, with topics ranging from ensemble Bayesian NN models for corrosion depth prediction over consequence modelling with incorporated SHAP-based interpretability and probabilistic uncertainty bounds [82], GBM/ Particle Swarm

Optimization-Tuned Feedforward ANN enhanced by Principal Component Analysis (PCA), linking corrosion growth to explicit reliability curves [83], physical integration of Bayesian NN (BNN) and Geographic Information System (GIS) [84], and theory-guided RF-metaheuristics [85, 86]. These approaches greatly improved the predictive capability of traditional tree ensembles, but neither demonstrated broad-spectrum applicability in terms of corrosion morphology. However, micro-scale consequence modelling [87] was traded off against network-wide integration of environmental and population exposure to provide higher strategic value, yet without historic validation. Subsea defect-interaction modelling [88] extended corrosion prognosis beyond single-defect assumptions in codes, while remaining geometry-specific with a gap between methodological complexity and applicability (Table 4).

Multi-mode degredation modelling [89-92] showed that custom ML frameworks, such as Bidirectional Recurrent NN (BRNN), and Random Survival Forest, consistently outperformed traditional survival and classification tuned on classic algorithms, especially with censored or incomplete data. In addition, survival-based models [91] delivered better time-stability performance than classification-only frameworks [90, 92], whose temporal interpretability was often hindered by the lack of hazard variables or synthetic data grounding. Hybrid risk assessment for steel and composite pipelines [93, 94] demonstrated that both fuzzy-logic integration and ML-optimised BNs circumvented the problem of expert bias. However, they suffered from instability in entropy-based weighting and independence assumptions (NoisyOR). Event-specific models [95, 96] dominated in niche hazard domains and significantly reduced error compared to empirical baselines; nevertheless, their application was limited by the small sample sizes required for reliable extrapolation.

The most currently available studies lack a standard protocol, exhibit weak cross-domain generalisation, with limited interpretability for operational implementation. Data shortages are particularly severe for hazard-specific cases and risk models of polyethylene (PE) pipelines [93-96], as compared to corrosion-centric approaches [84, 85], which perform well in terms of prediction accuracy but rarely provide measures of uncertainty and lack sufficient validation from different field settings. Models predicting survival [91, 92] address temporal degradation; however, they are sparse, indicating a general lack of explicit treatment of lifespans. New gaps are arising due to the lack of combined hazard-load-material interactive models and the consideration of degradation as stationary, rather than time-varying, which hinders scenario flexibility and accurate lifecycle prediction. Without addressing these gaps, improvements in the precision of forecast models cannot result in durable, field-optimised pipeline infrastructure.

Table 4. Comparison of methods used for service life prediction and degradation modelling in pipeline systems.

Reference	Application focus	Method / Algorithm	Data (type, size)	Key performance (proposed vs. benchmark)	Major advantages	Notable novelty	Limitations
[84]	Buried gas pipeline / External pitting corrosion / Defect depth & risk prediction	BNN, BBN, GIS	Field, n = 259 corrosion samples; n = 92 failure tests	Testing accuracy = 90 % vs earlier BBN study (higher accuracy)	Integrates uncertainty prediction with spatial risk assessment	BNN uncertainty + BBN-GIS risk framework	BNN uncertainty analysis not detailed; Inaccurate soil parameters
[85]	Buried pipeline / Pitting corrosion / Max depth prediction	RF, HMOGWO	Field, n = 259 samples	MAPE = 0.006 vs XGBoost ≈ 0.25; R^2 = 0.999 vs XGBoost ≈ 0.70	Very high accuracy with improved interpretability	Theory-guided ML + multi-objective feature selection	Generalization to unseen corrosion scenarios untested
[87]	Gas transmissio	BBN, GIS, Power-law	Field, n= 31607	R^2 = 0.92 vs PHMSA	Merges BBN uncertainty with	Bow-tie + BBN-GIS for	Environmental model

	n / External pitting corrosion / Social & environmental impact	corrosion model	9 pipes (located in Alberta, Canada)	impact radius data	GIS spatial analysis	spatial consequence	needs historic validation
[88]	Subsea pipeline / Interacting corrosion defects / RUL	NN, FEA, LHS	Simulated (FEA), n = 266200; Field ILI (rate)	R^2 = 0.99 (NN) vs FEA; DNV std ~15 % more conservative	Quantifies defect interaction effects	LHS + FEA + ML for defect interaction	Limited to 32″ X65 pipeline; lacks generalizability
[89]	Dry gas pipeline / Metal loss / Life & defect class prediction	BRNN with subnetworks for missing data	Field ILI, n = 11358 anomalies	R^2 =0.9687-0.9987 & vs DT R^2 = 0.93	Predicts life with incomplete inputs	BRNN framework for missing-data handling	Specific model limits not detailed
[90]	Oil pipeline / Corrosion & fluid hammer / Safety status prediction	RF Bagging, SVM, LDA, ANN	Simulated, n = 5000 runs	Accuracy = 96.8 % (RF Bagging) vs 83.1 % (LDA)	High accuracy (up to 97 %) in simulation	Combines corrosion + fluid hammer failure modes	Based solely on simulated data
[91]	Gas transmission / Multiple failure modes / Failure age & survival prediction	Random Survival Forest, Cox, ANN, SVR, RF, XGBoost	Field & simulated, n = 6386 failures; 24564 censored	MAE = 4.835 (Random Survival Forest) vs 10.957 (Cox); R^2= 0.810 vs 0.191	Handles censored data accurately	ML-integrated survival analysis	Key variables (coating, soil) missing
[92]	Multiple failure modes / Failure year prediction	ANN (MLP)	Field, n = 12182 incidents	R^2 = 0.8; no benchmark reported	Uses large dataset for hazard analysis	Covers time-dependent and independent hazards	Cannot model random external events
[93]	Urban PE gas pipeline / Multiple failure modes / Risk status assessment	Fuzzy TOPSIS, Structural Entropy, Cloud Inference	Field, n = 163 failures; expert opinion	"Medium risk" confirmed on-site	Resolves ambiguity/randomness in risk assessments	New integrated framework for PE gas	Limited on-site residual sample size
[94]	Urban gas pipeline / Multiple failure modes / Risk assessment	BN + BPNN (Noisy-OR gate)	Field, n = 1352 accident reports	Failure prob = 0.059 vs 0.082 (expert BN)	Lowers subjectivity of expert BNs	ML-optimized BN CPTs with Noisy-OR	Relies on Noisy-OR assumptions
[95]	Gas transmission / Natural force damage / Cause prediction	XGBoost, RF, SVM, MLPNN, k-Nearest Neighbors(KNN), Boruta	Field, n = 81 incidents	Accuracy = 92.3 % (XGBoost) vs 74 % (KNN)	High accuracy for natural-force failures	Links climate data to failure prediction	Small dataset

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[96]	HBNG pipeline / Jet fire / Impact radius	BPNN, CNN, RF, CFD	Simulations (CFD), n = 288 conditions	MRE = 2.8 % (BPNN) vs 14.9 % (empirical)	Large error reduction vs empirical	First ML jet fire radius model for HBNG	Model limits not reported

5. Pipeline integrity assessment

Pipeline integrity assessment is shifting from prescriptive and code-based procedures to predictive, data-driven workflows that integrate computational accuracy and physical fidelity in predicting structural response and damage growth. Corrosion, mechanical damage, and stress-driven failure emerge as dominant concerns, with ML surrogates replicating complex, costly simulations, such as FEA and smoothed particle hydrodynamics (SPH), to deliver rapid, high-fidelity damage estimates. These methods are applicable to corrosion growth prediction, magnetic force coupling techniques for non-destructive stress estimation, and ML-empowered defect detection within the framework of modern non-destructive evaluation (NDE) techniques. In the following, surrogate models empowered by ML techniques are reviewed, followed by AI-assisted corrosion growth and pitting depth prediction. The subsequent subsections are related to the use of NDE empowered by ML techniques for stress assessment and defect detection/classification, with ML-enabled geohazard-induced pipeline integrity assessment closing this section.

5.1. *ML-based surrogate models*

Several studies utilized surrogate models to enhance structural response prediction and damage forecasting [97-110] (Table 5). Applications considering stress and reliability under geohazards [97, 98] demonstrated that hybrid optimization (LSBES-ELM-MCS, GA-BP NN) became favorable, achieving minor errors ($R^2 \geq 0.98$) compared to benchmark models, particularly when fed with high-fidelity FEM or SPH datasets.

Decision trees (DT) and Multilayer Perceptrons (MLPs) served as surrogates for predicting structural deformation and stability [99-101], reducing FEA runtimes from several hours to a few seconds without compromising accuracy. CNN-ANN integrations resulted in probabilistic impact localization with an accuracy exceeding 94%. These techniques demonstrate the potential for near-real-time assessment, although training data are usually insufficient and target specific seabed and impact geometries, with model generalization yet to be proven. In collapse strength and pressure prediction [102, 103], Gradient Boosted Regression Trees (GBRT) and RF-based models trained on large parametric or manufacturing-history FEM datasets yielded R^2 approaching unity, far outperforming empirical standards like Det Norske Veritas (DNV). Their strength lies in robust generalization across defect variants; however, the absence of experimental validation (GBRT) and the high sensitivity of specific algorithms (k-Nearest Neighbors (KNN)) questions their universal application.

Table 5. Comparative summary of ML-based surrogate models for structural response and damage prediction in pipelines.

Reference	Application focus	Method / Algorithm	Data (type, size)	Key performance (proposed vs. benchmark)	Major advantages	Notable novelty	Limitations
[97]	MVMS & failure probability in buried pipelines under PGD	LSBES-ELM-MCS (BES+Lévy+ SA-optimized ELM + MCS on FEA)	150 FEA PGD cases	R^2: 0.980 vs. 0.947 (BES-ELM) and 0.834 (ELM)	Integrates FEA surrogate with reliability analysis	Triple-metaheuristic ELM optimization for high-precision stress prediction	Ignores thermal/residual stresses; ideal soil assumed
[98]	Weld VM stress under landslide	GA-BP NN on SPH-FEM	100 SPH-FEM landslide cases	Max err: <5% vs. field; 11.67% vs. >25% polyfit	Dynamic SPH-FEM realism	API RP 581 matrix from predicted stress	Specific weld config; narrow

							parameter range
[99]	Seabed liquefaction & displacement under wave-current	DT Regression on FSSI-CAS FEA	74 FEA	Acc: >95% vs. >98% FEA; Runtime: 5 s vs. ~50 days	Much faster with minor accuracy loss	FEA-ML fusion for rapid liquefaction prediction	Few samples; site-specific
[100]	Mode, PX, η for buckle arrestors	MLP on nonlinear FEA; vs. RF, KNN, SVM	182 FEA buckling cases	R^2: 0.964-0.999	High reliability for flipping mode	First ML use for arrestor mode/efficiency	Large η errors <0.6; no direct η tests
[101]	Probabilistic impact localization using accelerometers	1D-CNN + ANN (ABC-SS)	Lab: 17460 impacts	Prec: 94% force, 98% depth; F1 >99% CNN	Accurate, uncertainty-aware with minimal sensors	First CNN/ANN + ABC-SS for impacts	Needs validation in varied field conditions
[102]	Collapse strength of corroded screen pipes	GBRT on parametric FEM	332 FEM cases	Diff: 3.97% vs. 6.54% FEM/exp	Strong generalisation to defect combos	First GBRT for corroded screen collapse	No exp. validation for corroded pipes
[103]	Collapse pressure of lined pipes	RF/MLP/KNN/SVM on multi-step FEM	86400 FEM cases	R^2: 0.9999 (MLP)/ 0.9997 (RF)/ vs. 0.9127 (DNV)	Outperforms DNV code	Large FE dataset of manufacturing history	KNN low precision; >20% max err
[104]	Burst pressure of corroded offshore pipes	RF-FEA-CNN on stochastic RF-FEA	1000 RF-FEA samples	R: 0.959 (RF-FEA-CNN) vs. 0.935 (FCNN); Runtime: ~3.5 (CNN) s vs. 34 h (FEA)	30000× faster safety assessment	Links CNN features to physical failure	Lower accuracy for extreme d/t ratios
[105]	Burst pressure of corroded O&G pipes	TGNN (MLP + PCORRC formula)	Lab/Sim: 453 samples collected from 13 papers	R^2: 0.9945 (TGNN) vs. 0.9551 (NN)	Accurate and robust on small data	Integrates empirical law into NN structure	Formula choice may bias results
[106]	Burst pressure of single-defect gas pipes	ANN + SHAP with physics features	Lab/Field: 193 tests + 80 incidents	R^2: 0.974 (ANN) vs. 0.852 (DT)	Superior to empirical, field-validated	Physics features + SHAP interpretability	Only single defects; small dataset
[107]	Burst pressure, multi-grade steel	Extra Trees (ET) + SMOTE	Lab/Sim: 454 → 2172 augmented	R^2: 0.99 (ET) vs. 0.98 (XGBoost)	Solves imbalance/ overfit	First SMOTE for burst pressure	Single, simple defect shapes
[108]	Corrosion rate and life of pipelines	MFO-LSSVM + LSTM-DNN	Field: 70 obs. → 1000 augmented using interpolation techniques	R^2 life: 0.9999 (LSTM-DNN) vs. 0.945 (linear regression)	Robust from small data	Two-stage pipeline from rate to life	Needs more extreme-condition tests
[109]	Weld pass/fail disposition	XGBoost + LIME	Field: 1671 weld records	Acc: 98.5% (XGBoost) vs. 95.1% (ANN)/93.7% (RF)	High accuracy, big cost saving	First XGBoost weld disposition	Defect information excluded

Techniques employed for estimating failure pressure of corroded pipelines [104-107], include physics-guided Temporal Graph NN (TGNN), RF-FEA-CNN hybrids, interpretable ANN+SHAP

frameworks, and data-augmented ET. These approaches provided considerable computational advantages (speed-ups of up to 30000×) [104] and achieved great accuracy (R^2 up to 0.9945) [105], incorporating both physics-based and empirical formulae that enhanced interpretability. Limitations are common: most models addressed single-defect geometries and relied on synthetic or, to some extent, augmented data, raising issues regarding complex field corrosion. For corrosion growth and remaining life prediction [108], the hybrid Moth-Flame Optimization-Least Squares SVM (MFO-LSSVM) + Long Short-Term Memory Deep NN (LSTM-DNN) technique yielded near-perfect life prediction ($R^2 = 0.9999$), reflecting the effectiveness of multi-stage modeling in time-dependent degradation. However, its accuracy under extreme operating conditions still needs to be addressed. Lastly, the weld integrity classification [109] demonstrated the practicality of ML for inspection optimization: the combination of Extreme Gradient Boosting (XGBoost) and Local Interpretable Model-Agnostic Explanations (LIME) achieved 98.5% accuracy, suggesting the potential for eliminating up to 91.8% of NDE procedures. While cost-saving and interpretable, it cannot specify defect type or size, thus requiring improvement.

A cross-comparison of the ML-based surrogates [97-109] reveals that dataset-known limitations and validation rigor are among the most common bottlenecks. Almost all frameworks suffer from dataset paucity and poor quality, whether due to the low number of FE simulation [99, 100], limited field records [106, 108], a small sample size or incomplete types of defects. The over-dependence on simulated or enhanced defect scenarios is more prevalent when applied to burst, collapse, and stress-reliability models [97, 104-107], in which FE-driven training improved accuracy metrics at the expense of deployment realism. There remains poor real-world validation for all types of predictive systems, except for a few cases involving large laboratory or field tests [101, 109], which are still scenario-limited. Technical readiness is also hindered by poor interpretability, only a few used SHAP [106] or LIME [109], and systemic negligence of uncertainty quantification, despite its being mandatory for probabilistic safety frameworks [97, 105]. Cross-domain generalization failures are frequent, indicating that specific models focused on a subsystem (such as single-defect burst pressure or site-specific liquefaction). Multi-sensor or multi-source integration, beyond impact localization [101], remains largely unexplored in most studies, lacking a clear regulatory or operational adoption pathway, which causes a gap between academic proof-of-concept and field-deployable decision support.

5.2. *ML for defect growth prediction*

Defects can grow over time and threaten pipeline integrity; therefore, the prediction of corrosion growth rate [111-116], corrosion depth [85, 117-122], crack growth rate [123, 124], and stress intensity factor (SIF) in combined crack and corrosion [125] is essential for the seamless functioning of pipeline systems. The recent progress achieved in ML-based predictions consists of three main categories (Table 6). The first group, metaheuristic-hybrid architectures employed evolutionary or swarm optimizers to fine-tune deep or ensemble learners, delivering R^2=0.960-0.999 and surpassing conventional baselines. These techniques include SSA-LightGBM [111], GAN with LSTM and Multi-Head Attention optimised by GWO and SVR (LAGAN-GWO-SVR) [113], CNN-BO-Transformer [114], Radial Basis Function NN enhanced by PCA [115], LSTM with transformer [116], Marine Predator Algorithm with Self-Adaptive Differential-Biased Optimization and Bidirectional Long Short-Term Memory (MSDBO-BiLSTM) [117], XGBoost [118, 126], hybrid Four-Vector Intelligent Metaheuristic [127], theory-guided techniques [128], Hybrid Multi-Objective Grey Wolf Optimizer (HMOGWO)-RF [85], and Improved Grey Wolf Optimizer-tuned SVM (IGWO-SVM) [119].

The second group, tree-based and physics-integrated models, includes RF [112], Bayesian-RF/ Categorical Boosting (CatBoost) [120], and Physics-ML hybrids [121], together with hydrogen-assisted fatigue crack growth (HA-FCG) studies employing CatBoost [123] and ANN/RF [124] frameworks, which combined statistical inference with crack growth mechanisms (e.g., Paris law) to link crack-growth rate (da/dN) responses under varying hydrogen pressure to predictive modeling accuracy. While the majority of studies focused on either corroded or cracked pipelines, an XGBoost model [125] predicting

Table 6. Comparative analysis of ML models for corrosion growth and pitting depth prediction.

Reference	Application focus	Method / Algorithm	Data (type, size)	Key performance (proposed vs. benchmark)	Major advantages	Notable novelty	Limitations
[111]	Average corrosion rate for buried pipelines	SSA-LightGBM with SPDA augmentation	Field; 300 (augmented from 60)	SSA-LGBM: R^2= 0.9813 vs. LGBM: R^2 ↓ 11.41%	High accuracy with small set; SSA tuning	First SSA-LightGBM for corrosion rate	Omits steel type/protection; no life estimate
[112]	Buried steel pipe corrosion rate prediction.	RF regression vs SVR/kNN/CART.	Field coupons; 78 samples.	Train R^2 0.98 / MAE 0.0058; Test R^2 0.56 / MAE 0.0358; SVR worse.	Captures non-linear direct stray current effects.	Uses negative E_{off} shift as predictive feature.	Large overfit gap; very small dataset
[113]	Internal corrosion rate prediction in oil and gas pipelines	LAGAN-GWO-SVR (GAN + LSTM + MHA + GWO-optimized SVR)	Field; 67 samples (7 wells) → augmented to 1054	R^2 0.982 (LAGAN-GWO-SVR), vs SVR (R^2 0.749)	High accuracy, strong generalization, robust parameter optimization	LSTM-MHA-integrated GAN for non-IID corrosion data; combined with GWO for automated SVR tuning	Real data scarcity; interpretability and cross-material validation
[114]	Internal corrosion rate (mm/yr)	Hybrid CNN-BO-Transformer; OLGA inputs	Lab/Field; N=2556	R^2=0.9807 (CNN-BO), vs 0.8025 (SVR)	High accuracy, interpretable (DeepSHAP)	Combined CNN-BO with Transformer	Sparse Cl^- data; few low-rate samples
[115]	External corrosion rate (mm/a)	PCA + optimised Radial Basis Function NN; UT gauge	Field; N=200; acidic/saline soils	Acc=91% (Radial Basis Function NN), +30-50% prediction reliability vs CSEM/CNN	Stable across complex soils	PCA-enhanced Radial Basis Function NN	Omits seasonal/dynamic factors
[116]	External corrosion; rate/max pit depth	Hybrid Transformer-LSTM	Field (China N=78; Mexico N=193)	MAE (Transformer-LSTM) ↓85.5%, vs LSTM	Robust; low-data stability; low training time (14 s)	Transformer-LSTM for corrosion	Needs structural/hyperparameter tuning
[117]	Corrosion depth in buried pipelines from NDT data	MSDBO-BiLSTM; TEM + lab data, dung-beetle metaheuristic	Hybrid lab + simulation; size NR	MSDBO-BiLSTM: R^2= 0.9642 vs. RBF: R^2 = 0.8227	Higher accuracy/stability; addresses slow convergence	First MSDBO-BiLSTM for TEM-based corrosion depth	Simulated data in air; limited soil validation
[118]	Max. pitting depth in underground pipelines	DE-XGBoost (XGB-FS), SVR, ELM, MARS	Field; 259 obs.	XGB-FS: R^2= 0.8407, vs. SVR-FS: R^2= 0.7518	Improves accuracy/generalization via DE tuning + FS	Joint hyperparameter + feature optimization with DE	Larger errors for severe pits; costly optimization
[85]	Max. pitting depth with theory guidance	HMOGWO-RF, physics-informed	Field; 259 samples	HMOGWO-RF: R^2= 0.999, RMSE = 0.055 vs. RF: RMSE ≈ 0.08	Extreme accuracy/stability; multi-objective balance	Theory-guided multi-objective RF optimization	No validation for different operating conditions
[119]	Max. pitting depth for strength eval.	IGWO-SVM; Pearson FS	Field; 255 samples	IGWO-SVM: RMSE = 0.52 mm, MAPE = 0.10 vs. RF: RMSE = 1.45 mm	Outperforms 8 benchmarks; robust FS + tuning	IGWO-SVM with correlation-based FS	Small set; excludes inspection data
[120]	Max. pitting depth in	Bayesian-RF/CatBoost	Field; 245 samples	CatBoost: 0.5924; RF: R^2= 0.5915, vs. DT: R^2= 0.0171	SHAP + permutation	Bayesian-optimized tree models with	Severe overfit; small set

	buried pipelines				importance insights	interpretability	lacks deep-pit cases
[121]	Int./Ext. pitting corrosion depth in water pipelines	Physics-ML (power-law + RF /XGB residuals)	Field; 377 ext., 259 int.	RF-HCP(ext.): R^2= 0.958, vs. ANN: R^2= 0.832	Trend separation enhances accuracy	Physics-inspired hybrid residual correction	Limited genelizability beyond dataset; avg-time internal pits
[122]	Max. pitting depth in buried pipelines	DNN (Generalization, Gen-Mem)	Field; 259	Gen: RMSE = 0.0610 mm vs. SVR-FFA: RMSE = 0.2909	Resolves hybrid overfitting; high accuracy	Novel Gen/Gen-Mem DL arch.	Overestimating max. pitting depth
[123]	HA-FCG-crack-growth rate	CatBoost regression	Experimental + literature data: 47	CatBoost outperformed XGBoost, ANN, SVM, RF, KNN, DT	Captures nonlinear HA-FCG behavior with limited data	Sequential modular ML architecture using RA/FT enhance interpretability and generalization	Unrealistic frequency; limited field datasets
[124]		ANN and RF regression	400 data from literature	Proposed ANN: R = 0.98 (test, X70) Best RF: R = 0.9983 (test, X52), R^2 = 0.9963	ANN→ superior interpolation, and smooth trend	ANN/RF application to HA-FCG prediction with derivable closed-form polynomial relation for design use.	Limited datasets for load ratio and frequency; lower accuracy at very low da/dN
[125]	SIF prediction for combined crack-corrosion	XGBoost	385 samples (FEA)	R^2=0.987	Defect-interaction impact on life reduction	First integration of XGBoost-predicted SIF	Case-specific crack - corrosion; pressure fixed

SIF of combined crack and corrosion defects extended ML application to coupled degradation scenarios.

The third group, pure DL architectures, notably the Generative Memory DNN [122] aimed to tackle overfitting while maintaining low prediction error. Hybrid techniques tended to enhance convergence stability and robustness; nevertheless, they caused heavy computational cost and often relied on small or augmented datasets. Tree-physics hybrid models enhanced interpretability but were prone to losing predictive robustness in severe or non-standard pit corrosion profiles. Standalone deep networks achieved competitive error reduction without metaheuristics but faced black-box limitations and lacked extensive validation.

Techniques employed in this section face the majority of common challenges, including scarcity and quality issues in training data, a lack of standardized benchmarking protocols, poor cross-domain generalization, and the absence of uncertainty quantification, shortcomings hindering the effectiveness of ML techniques as core functionalities in real-world applications. These models did not achieve interpretability to a vast extent, as explainability is primarily limited to physics-integrated and tree-based methods. Computational scalability barriers prevent scaling in practice due to complexities of multi-sensor fusion. Two gaps emerging from this synthesis are the limited integration of temporal degradation modeling being necessary for long-term corrosion prediction, and vulnerability to overfitting on small or balanced datasets. These challenges will demand multi-source field data standardization, physics-interpretable ML couplings with robustness to environmental variabilities, benchmarking frameworks that contextualize accuracy within uncertainty, and validation protocols that acknowledge operational, regulatory, and cross-domain transfer considerations.

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5.3. *ML-NDE applications*

Several techniques developed for NDE-ML research in pipeline integrity applications enhanced detection or predictive capabilities via sensing-algorithm coupling [112, 129-136] (Table 7). The Cross-Domain Generative Diagnosis Method (CDGM) utilized Denoising Diffusion Probabilistic Models (DDPM) to enhance the classification of wall thinning and weld defects, albeit at the expense of reduced generation speed [129]. A multi-coil Strengthened CNN (TWSC) framework, fusing Short-Time Fourier Transform (STFT)/ Continuous Wavelet Transform (CWT) spectra into a strengthened CNN architecture, delivered high accuracy for pitting and crack diagnosis on challenging, low Signal-to-Noise Ratio datasets; nonetheless, its defect coverage remains narrow [130]. The spatial magnetic signals extraction-RF inversion framework [131] enabled inner-surface crack sizing with moderate RMSE ; however, it was restricted to regular, single-surface defects. In contrast, crack identification and sizing using Magnetic Flux Leakage (MFL)-KELM (RBF kernel) achieved high accuracy with fewer samples, tanks to RBF-kernel integration into the model physics [132]. Furthermore, magnetoelastic coupling combined with RFR [133] estimated pipeline wall stress under concurrent pressure, temperature, corrosion, and material variations, advancing magnetic-mechanical learning beyond conventional MFL-based defect sizing. More recently, a task-oriented physical collaborative network enabled joint defect detection and size quantification from MFL signals by embedding domain knowledge and cross-task feature interaction, achieving stable performance on complex inspection scenarios [137].

Physics-guided 3D shadow modeling (3D-SM) integrated into CNNs enabled corrosion crack localization with strong recall; however, its susceptibility to environmental noise may obscure minor defects [134]. RF regression, applied to electrical potential monitoring under direct current stray current conditions, predicted corrosion rate with suitable accuracy; however, it exhibited overfitting due to the limited field data [112]. Common bottlenecks include dataset scarcity, lack of standardized benchmarks, limited interpretability, and weak real-world validation, while revealing additional gaps, including domain-specific augmentation optimization [129] and physics-data coupling exploitation [134].

5.4. *Geohazard-induced pipeline integrity*

Geohazard-inducing pipeline integrity has progressed by developing physics-informed and hybrid ML techniques [138-143], including Physics-Informed Neural Networks (PINNs)-based Permanent Ground Displacement (PGD) modeling [138, 139], hybrid GIS/remote sensing with ensemble ML for landslide susceptibility and vulnerability [140], multi-sensor XGBoost models for landslide-induced stress prediction [141], FEM-coupled white shark optimization-ELM analysis of blast vibration effects on HDPE joints [142], and fast-running Feedforward NN metamodels for seismic PGD strain estimation [143] (Table 8). These techniques improved generalizability, computational efficiency, enabled scalable hazard-response mapping, and employed sensor-integrated data to enhance situational awareness.

PINN-based techniques [138, 139] delivered mesh-free, physics-consistent predictions, but with a relatively high training cost, while remaining largely constrained to elastic, isotropic or simplified nonlinear PGD scenarios. Landslide-related models [140, 141] demonstrated hazard-asset coupling and limited multi-sensor integration, yet relied on small, sometimes noisy datasets and lacked standardized performance metrics for strain/displacement outputs. FEM-ML couplings explicitly developed for a particular type of hazard [142, 143] achieved great predictive accuracy and computational efficiency compared to pure FEM. Despite remarkable progress, several bottlenecks remain: black-box architectures and single-site/small-scale datasets hinder scalability. Among the long-standing gaps, we can identify the absence of uncertainty quantification, inadequate real-world multi-event benchmarking, neglect of hazard interaction effects, and a lack of regulatory acceptance.

6. Pipeline inspection and maintenance planning

For cost-effective pipeline integrity management, inspection and maintenance planning is now increasingly leveraging advances achieved by AI and ML. AI and ML have the potential to integrate data of diverse types from inspection, environment, and operation into a unified decision-support framework.

Table 7. Advanced defect detection and classification approaches integrating ML with NDE techniques for pipeline integrity assessment.

Reference	Inspection focus	Technique / Algorithm	Data	Performance (Proposed vs. Benchmark)	Advantages	Novelty	Limitations
[129]	Wall thinning and weld defect detection	Magnetic eddy current, X-ray; CDGM (DDPM+U-Net) with VGG16/DenseNet evaluators	Field, lab, public; 634 signals	82.16% vs 71.80% (orig.) / 78.23% (DDPM); Cdge-Score 0.66 vs 0.23; FID 149.1 vs 155.96	Cross-domain augmentation lifts accuracy by ~13 pp	Collaborative DDPM training across field & simulated data	Slow sample generation; limited speed scalability.
[130]	Five-class defect ID (pitting, cracks)	3-coil bidirectional eddy + MFL; TWSC (STFT+CWT fusion, strengthened CNN, transfer learning)	Field/lab; 371 samples; Signal-to-Noise Ratio 30-50	96.0% vs SVM 93.3% / CNN-STFT 86.7%; F1 96.1%	High accuracy on small-noisy datasets	Fusion of multi-time-freq features with strengthen-conv & transfer learning	Small dataset, narrow defect coverage
[131]	Inner-surface crack sizing	Spatial Magnetic Signals Extraction + RF; FE	FE; 298 samples; depth 0.1-0.5 mm, width 0.4-1.4 mm	R^2 0.801 (RF) vs. 0.705 (DT)	Low-mag avoids tool jamming	Novel Bx-By features for inversion	Depth less accurate; inner-surface only
[132]	Defect classification and sizing	MFL + KELM (RBF kernel)	FE simulations; 223 MFL signals	Accuracy: Classification 98 %, Regression ≈ 99.8 % (vs. ELM 90 %)	High-accuracy detection; adaptable to multi-defect forms; hybrid FE-ML coupling	Integration of RBF-kernel ELM with physics model	Lower accuracy with single-feature input; no field validation
[133]	Stress under multi-factor loads	Magnetoelastic + RF; FE	FE; 1188 models; varied P/T/corrosion/material	R^2 0.870 (RFR) vs. 0.627 (SVR);	Accurate with missing inputs	Multi-ML comparison; direct stress from flux	Static-load assumption; high-T error
[134]	Corrosion crack detection & localization.	Optical 3D-shadow modeling + CNN	Public field images; 900 (700/200)	93.53% acc.; Recall 92.0; Benchmark NR	Physics-based features enable micro-crack detection	Integrates engineered 3D-SM features into CNN	Noise can mask small cracks

Recent studies have used physics-informed and/or data-driven approaches to identify hazardous sections and optimize inspection time intervals. For instance, researchers have combined GIS and ML for fault location hotspot detection and utilized Light Detection and Ranging (LiDAR)-assisted surface inspection to predict faults. These innovations not only improve detection precision and timing but also enable adaptive maintenance planning. Adaptive planning addresses evolving threats throughout the pipeline lifecycle, ultimately enhancing operation safety, cost efficiency, and regulatory compliance. In the following, the focus is on the application of AI or ML techniques in condition monitoring, corrosion rate prediction, and spatial and inspection prioritization.

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Table 8. Summary of physics-guided ML approaches for geohazard-induced pipeline integrity.

Reference	Geohazard / Focus	Technique / Algorithm	Data	Performance (Prop / Bench)	Efficiency	Key advantage	Novelty	Limitation
[138]	PGD; elastic displacement-strain in buried metallic pipelines	PINN + transfer learning	22000 collocation points; FEM/FDM, Latin Hypercube Sampling (LHS)	Axial displacement MSE: 10^{-9}-10^{8} m; Lateral displacement MSE: 10^{-6}-10^{5} m	Train: 4-39 min (PINN) vs. FEM 138.89 h (all samples)	Mesh-free, scalable PGD mapping; strong generalizability	Single PINN covers continuous hazards	Elastic only; Intact pipe
[139]	PGD; elastic displacement-strain in buried CFRP pipelines			Tensile strain relative MSE: 2.76% unreinforced, 3.86% reinforced	Train: 0.51-1.42 h (PINN) vs. FEM 15 s (each sample)	Meshless nonlinear PGD predictor	PINN on semi-explicit PDEs	Isotropic assumption for CFRP; wall buckling neglected
[140]	Landslide susceptibility + pipeline vulnerability	GIS, RS; RFE-PSO-AdaBoost + FCM-CRITIC	917 land-slides; 124.4 km pipe; 3986 segments	Acc: 0.8255; F1: 0.8389; AUC: 0.8821	NR; scalable to networks	Objective hazard-risk scoring	Spatial hazard-asset hybrid model	Static; limited failure data
[141]	Landslide stress in steel pipes	Strain, FBG, inclin., GNSS + XGBoost	400 sets; 4 sites	R^2: 0.9893 (XGBoost); 0.6233 (SVR)	NR; fast via parallel computing	Works if strain gauges fail	Multi-sensor landslide→stress link	Small database; noisy labels
[142]	Corrugated HDPE; Blast vibration	Sensors + White Shark Optimization-ELM; FE coupling	40 samples (site + FEM)	R^2 RPPV: 0.9969 / 0.8461; R^2 RPPD: 0.9833	FEM > 24 h; ML time NR	↑ Blast-response accuracy	First White Shark Optimization-ELM for HDPE	Single-site; geology untested
[143]	Seismic PGD; tension/comp. strain	FE data; Feedforward Neural Networks	108 FE simulations	R^2: 0.99897 (BR)	≪ FEM time	Rapid design/route optimization	Decoupled ANN metamodel	Small database; black-box

6.1. *ML-assisted condition monitoring*

Smart sensing and ML inspection techniques have substantially contributed to pipeline monitoring as summarized in Table 9. The techniques used include those for crack detection [144-147], quantification of coating loss [148], leak detection [149, 150], and determination of deformation [151, 152]. Studies utilized various sensing tools, including LiDAR (laser-based) [144], Closed-Circuit Television (CCTV) [145], acoustic emission (AE) sensors (sensing sound from cracks or leaks) [149], endoscopic profiling (TV inspection) [151], and PIG data (a machine inspection device) [152]. These sources were processed using other algorithms, such as anomaly detection with isolation forests [144], deep image models with Swin Transformers [145], a decision-making process informed by DRL [146], and a prediction process handled by GBM [152]. Most were reported to achieve higher accuracy than previous benchmarks; for some, the classification accuracy even exceeded 99%, and they could do so under noisy conditions. Among the innovative aspects are multi-stage hybrid processing and domain adaptation (bootstrapping models to new data sources). Some studies [144, 145, 148, 152] integrated hardware and software for real-time use, benefiting from efficiency gains (via parallelism and denoising) that accelerated both training and inference.

Table 9. Comparative analysis of smart sensing and ML for pipeline anomaly monitoring.

Reference	Inspection focus	Technique / Algorithm	Data (type, size)	Performance (proposed vs. benchmark)	Efficiency	Advantages	Novelty	Limitations
[144]	Cracks on inner pipeline surface; detection, classification (position, angle, depth)	LiDAR (TF-Luna, 360° scan) + Isolation Forest + XGBoost	Lab; NR; 10 cm radius pipe, 1 cm crack width	Proposed: Acc 93%, F1 0.93; Benchmark (CNN): Acc 91.8%, F1 85.34%	Cost-efficient, scalable; remote monitoring via ESP32/mobile app	Dual unsupervised/supervised ML reduces false positives	Integration of LiDAR with dual-ML framework	Single crack size, lab-only validation
[145]	Welding cracks, infiltration, branch joints; detection/localization	CCTV + Swin Transformer in Faster R-CNN with FPN.	Public field dataset; 1252 images; CCTV inspection.	mAP 0.903, Recall 0.933; Benchmark Faster R-CNN mAP ~5% lower	Designed for real-time detection	High accuracy for small-scale defects	Swin Transformer within Faster R-CNN for defect detection	Slight loss for rare defect categories
[146]	Cracks, corrosion, welds; automated anomaly detection & classification.	Eddy Current Testing array + DRL with domain adaptation.	Lab and field; 183 pull-in, 91 field anomalies; high-noise environment.	Field: Recall 97.4 %, Precision 94.9%; Benchmark (KNN): Recall 65%	Adaptable for remote/ILI via pre-training/fine-tuning.	Robust in noisy, real-world settings.	Deep RL with hierarchical reward and spatiotemporal exploration	No defect sizing; sensor jitter can misclassify
[148]	Coating loss; classification of 4 defect types	Hybrid eddy current/MFL detector + GA-SENet-ResNet18 (wavelet TFRs)	Lab; 250 samples → 998 augmented; 89 mm steel pipe	GA-SENet-ResNet18: Acc 96.83%, F1 96.68%; ResNet18: Acc 94.77%, F1 94.31%	Training faster than benchmarks	Signal denoising + attention + GA tuning.	GA-SENet-ResNet18 with VMD-DWT denoising	Lab-only; no field validation
[149]	Offshore leaks; detection/classification in noisy conditions	AE (VMD + GBDT)	Hybrid lab/field; 1800 samples; marine noise fusion	VMD-GBDT: Acc 99.44%; Benchmarks: SSA-CNN 96.3%, SA-PSO-BPNN 93.7%, WT-SVM 84.8%	Less complex than benchmarks	Robust (>99% acc.) under strong noise	VMD feature extraction + GBDT classification	AE defect identification limitations
[150]	Leaks from cracks/ruptures; “leaky” vs. “non-leak”	AE + DBN-GA-LSSVM on CWT scalograms	Lab; 960 samples; stainless steel pipe; high pressure	Proposed: Acc 99.66%; CNN: 96.95 %; CNN-CAE 96.72%	Feature extraction + classification in ~920 s	High accuracy via ELIS scalograms and optimized DL	Enhanced CWT scalograms with hybrid DL	Lab-only; no localization; private data
[151]	Dents, holes, slits; inline defect detection/classification	Endoscopic laser profiling + hierarchical bilinear pooling (HBP) DL with	Lab; 2160 train, 539 test images; various plastics	HBP: Acc 98.33%; CNN-TL: 89.80%; CNN: 83.85 %; SVM: 72.17 %	Instant diagnosis; adaptable hardware	Robust to scan/laser variations	Endoscopic laser profiling with pre-trained HBP network	2D imaging cannot measure depth

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[152]	Dents, offtakes, ovality; classification to cut false alarms	transfer learning PIG sensor data + SVM & GBM	Field data from 26 PIG inspections	TPR 100%; Legacy system false alarms −95%	Parallel processing in Hadoop software	Maintains 100% detection while cutting false alarms	Parallel ML processing of imbalanced PIG data	Only covers 3 defect types

Despite the advantages, cross-comparisons expose key deficiencies that limit operational application. Most studies do not generalize well across domains, lack uncertainty estimation, and use a fixed evaluation protocol. These issues stem from the limited availability of datasets, a lack of diverse field conditions, and a focus on subsystems, with real-world validation missing in almost half of the studies. Technical limitations, such as inability to size defects [144-146, 151, 152] or achieve computational scalability [146, 150, 152], also prevent use for comprehensive integrity management. Chhotaray and Kulshreshtha [152] addressed practical approaches, but regulatory and deployment gaps remain. This highlights the need for rigorous, multi-sensor, interpretable, and field-diverse AI- or ML-assisted models validated in standardized, scalable, and context-rich settings.

6.2. *Maintenance-oriented defect assessment*

ML models in this subsection [153-159], including Gradient Boosting Regression (GBR) [153], Physics-guided D-ResNet [154], Hybrid Meta-Learner Ensemble [155], Hybrid WA-SRF [156], GNN + Graph Attention Network (GAT) [157], RF/GB [158], and Hybrid WOA-XGBoost [159] support maintenance scheduling, resource optimization, and strategic risk mitigation over long operational time intervals (Table 10), as compared to defect models in subsection 5.2 which focused on mechanistic, rate-based prediction models. The majority of studies covered a variety of corrosion-related topics, including the perforation risk of internal corrosion [153], pitting severity [154] and multi-class risk-based categorization of corrosion [155], with some presenting general defect or failure prediction frameworks [156, 157]. In addition, the risk of failure in cracked pipelines has been predicted through ML techniques [158, 159] (Table 10).

Interpretability has advanced with SHAP-driven analysis [154, 159], addressing issues related to black box feature of machine learning models. Active learning [153] has reduced reliance on inspection data in later stages, maintaining accuracy. Meta-Learner ensemble [155] exhibited enhanced stability and class-balancing, whereas domain-integrated methods, such as Wombat Algorithm-driven Scalable RF [156] and Graph Neural Network (GNN) + GAT [157], have demonstrated reliable model performance with large or multi-source datasets. However, multi-sensor data fusion remains under-represented, with notable integration only in Ref. [153]. Interpretability was addressed in part [154], but most frameworks are opaque in practice. The scalability was highlighted [156], and the modelling possibilities for corrosion growth, key to predictive maintenance, were missing in the reviewed literature. Complete coverage of environmental conditions was also lacking, with extreme climate and soil chemistries being ignored. Thus, even though the works demonstrate a diversity of approaches and some clear improvement over traditional methods in predictive performance, there is a lack of published works on bridging this gap by translating these powerful methodologies into deployable, field-friendly, regulatory-compliant, and environment-tolerant solutions.

6.3. *ML-driven spatial and inspection prioritization*

GIS spatial analysis was combined with ML to enhance pipeline inspection targeting decision-making [160-165] (Table 11). High-accuracy hotspot mapping [160] and multi-source risk fusion [161] both hinted at the promise of integrating spatial statistics, ensemble learning, and physical models for cross-hazard prioritization; however, they struggled with sparse spatial-temporal sampling, underreported incident information, and the absence of cost-benefit coupling. Weld inspection ranking [162] and

Table 10. Maintenance-oriented growing defect prediction.

Reference	Corrosion focus / aim	Technique / ML	Data source, size	Perf. metrics; proposed vs. baseline	Advantage	Novelty	Limitations
[153]	Internal corrosion and perforation risk; mean metal loss/km	GBR in AL; sensor fusion; ILI (MFL)	Field (offshore); N=360 ILI	R^2 from 0.803→0.925 after 3 AL iters	Cuts ILI training sample needs	First AL use for ILI	Poor raw data quality; extrapolation limits
[154]	Pitting risk	Physics-guided D-ResNet + KPCA; SHAP	Field; small; coastal T, humidity, salt fog; 11 vars	MAE=0.4616 (D-ResNe); 0.775 (SVM)	High-precision, interpretable risk	D-ResNet + SHAP integration	Limited to temperate onshore; no deep-sea/Arctic test
[155]	General corrosion; risk class (H/M/L)	Hybrid Meta-Learner Ensemble (8 models)	Open-source; N=10293	Acc=94%, Prec=0.93 (High); >RF, LR	Balanced class performance	First meta-learning ensemble for risk grading	No data source type; real-time monitoring needed
[156]	General failures included, corrosion/leakage	Hybrid Wombat Algorithm -SRF	100 (Kaggle);	Acc=98.7%, >SVM (97.4%), SA-BP-PSO (93.33%)	High accuracy	WA-optimized RF	Affected by real-world sensor noise
[157]	General defects/leakage	GNN + GAT; FE-coupled (ANSYS); sensor fusion	Field/Sim; N=5000	Acc=92% (defects), 85% (leakage); +15% vs traditional	Multi-source data integration	GNN aggregation of spatio-temporal data	Data collection hindered by environment
[158]	Failure mode classification in cracked pipelines	Five ML models (NB, SVM, DT, RF, GB); RF/GB best	250 full-scale burst tests (lab)	Accuracy > 95 %; baseline 55 %	Outperforms mechanics models by 40 % margin	ML comparison for failure mode classification	Limited generalization; black-box deterministic output
[159]	Crack-in-dent (CID) failure risk	Hybrid WOA-XGBoost optimized by WOA	Parametric FE (XFEM/ABAQUS); N = 132	R^2 = 0.986; + 31.7 % vs RF; +3 % vs unopt. XGB	High accuracy and robustness; interpretable via SHAP	First interpretable FEA-ML hybrid for CID failure pressure	SHAP partly captures feature interaction; limited case transferability

Table 11. GIS-ML and spatial intelligence for pipeline inspection and decision support.

Ref.	Focus / Aim	Technique / ML	Data	Perf.	Advantage	Novelty	Limit.
[160]	Multi-hazard; rank high-risk segments	Spatial (Gi*, Moran's I) + RF/XGB/GB/LR; net-aware weights	2971 incidents	RF 82 % gas; 94 % oil	Pinpoints hazard clusters	Spatial-ML integration with net structure	Under-reported minor cases; no cause explainability
[161]	Multi-hazard risk; strategy prioritization	GIS + ML/physics (SVR, RF; diffusion/leak models); multi-source	NR	SVR MAPE 13.56 %; R^2 0.37	GIS-ML-physics risk fusion	Adds situational awareness; Rough Set fault tree simpl.	Low R^2; fixed environment factors
[162]	Weld defect inspection ranking	RF + CatBoost; NDE inputs; SHAP	675 cases	Acc 0.836 (RF + CatBoost); 0.601 (RF)	Boosts weld-inspection accuracy	First RF + CatBoost weld model with SHAP	Data-quality dependent; high training cost

[163]	Operational state recognition (valves, pigging)	RF/AdaBoost/SVM; time-graphlet feats	Field (SCADA); 9 stations; 2 mo.	Acc 98.9 % (AdaBoost) vs 91.4 % (TSF)	Timely op-state alerts	Graph feature + shapelet mix	One station; univariate only

operational state recognition [163] demonstrated that focused ML models can improve detection accuracy and situational awareness; however, both lacked scalability and generalization, being limited to a narrow subsystem scope. Other deficiencies include weak cross-domain generalization and inconsistent operational uptake, along with the absence of uncertainty quantification. Meanwhile, interpretability shortfalls and exhaustive evaluation designs impede operational deployment. To bridge these gaps, in particular, multi-sensor integration, robust validation across diverse field conditions, and formal linkage to regulatory or economic decision-making frameworks are essential for translating GIS-ML intelligence into sustainable pipeline integrity management.

7. Discussion and future outlooks

Herein, a comprehensive quantitative analysis of the studies presented in sections 4-6 is presented, comparing high-performance ML techniques, the key variables used in these ML architectures, the most frequent research topics, and persistent research gaps.

7.1. *High-performance ML techniques*

The cutting-edge ML techniques for pipeline reliability, integrity, and inspection have been summarized in Figure 6-7, and Table 12. These include hybrid methods, physics-informed signal

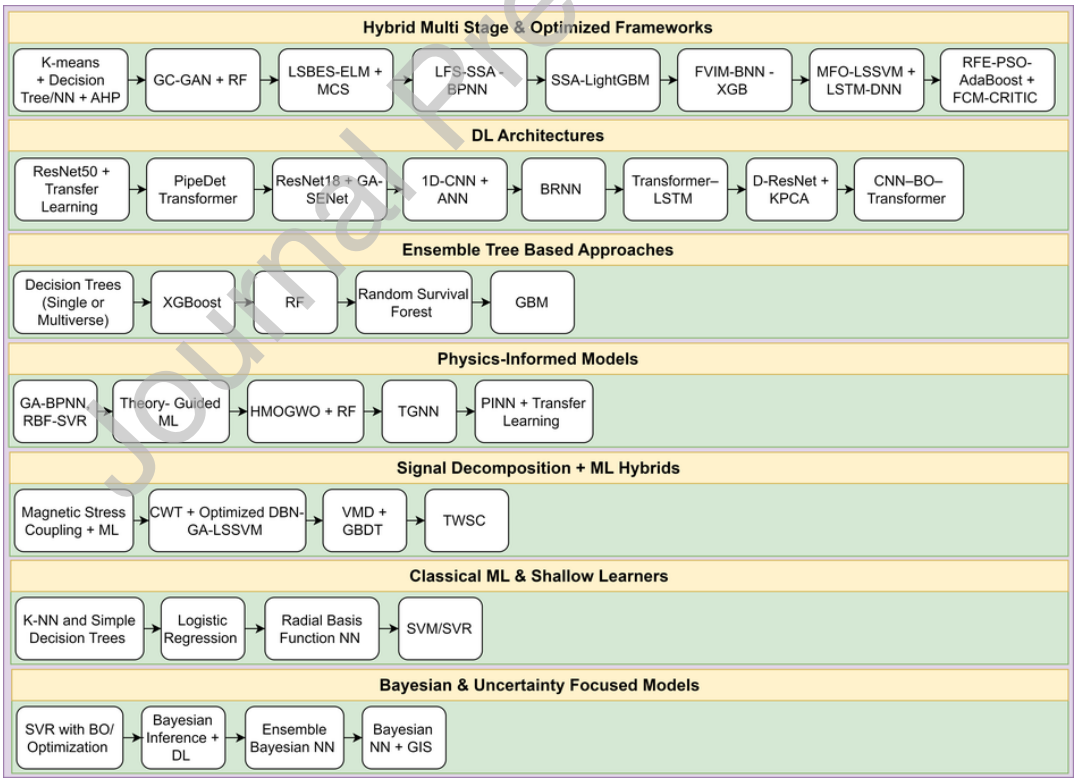


Figure 6. Classification of ML techniques for pipeline applications based on methodological complexity.

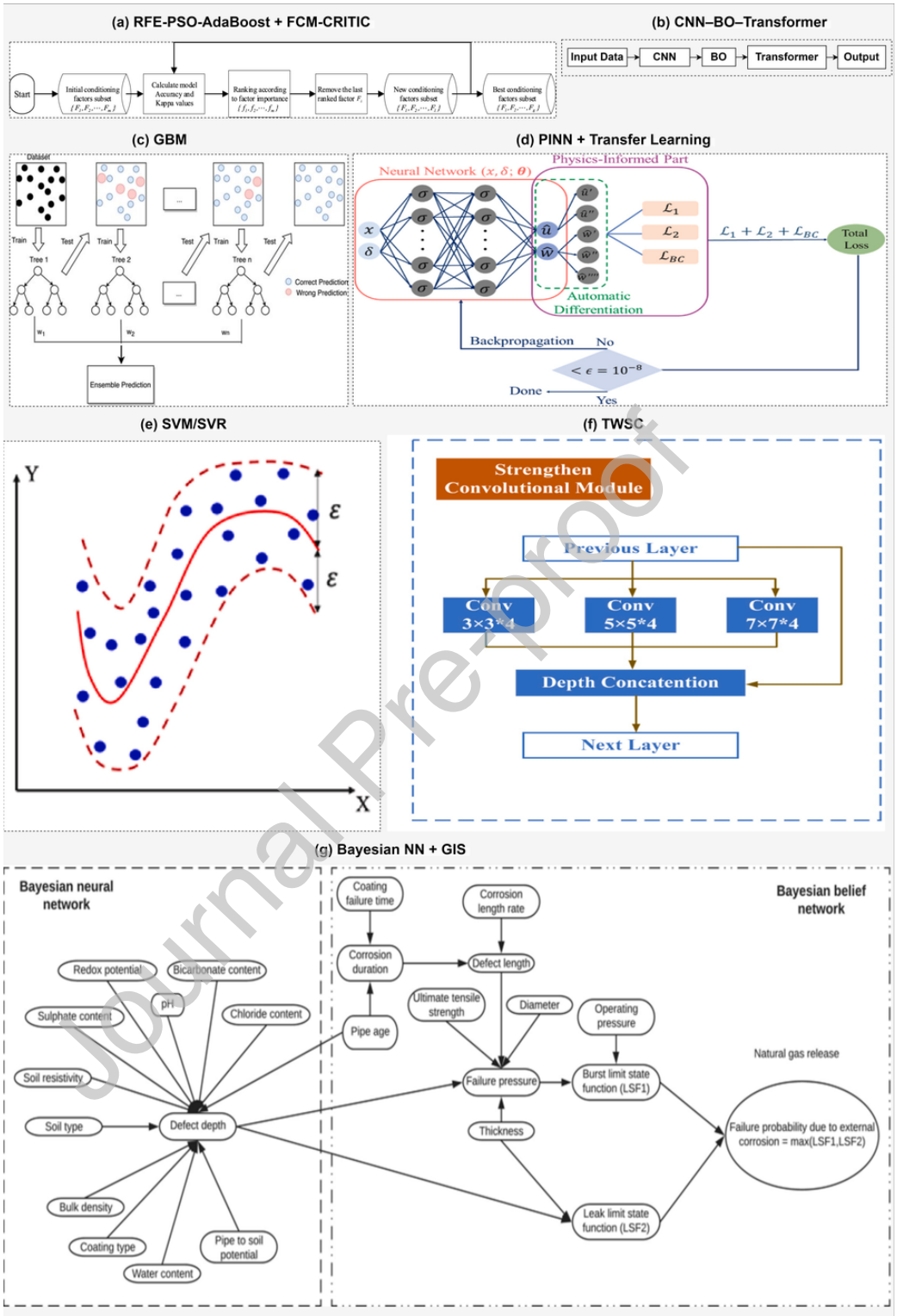


Figure 7. Examples of high-performance ML models applied in pipeline studies: (a) RFE-PSO-AdaBoost + FCM-CRITIC, (b) CNN–BO–Transformer, (c) GBM, (d) PINN + Transfer Learning, (e) SVM/SVR, (f) TWSC, (g) Bayesian NN + GIS (Adapted and redrawn from literature by the authors [84, 114, 138, 140, 166, 167]).

Table 12. Summary of high-performance ML techniques applied to pipelines for reliability-based design, integrity assessment, and inspection/maintenance.

Theme	ML technique	Category	Data modality	Advantages	Limitations	References
Reliability and risk assessment	LFS-SSA-BPNN	Optimized NN	FE simulation + MCS	High stability and accuracy	Single seismic wave, fixed failure criterion	[65]
	LSBES-ELM + MCS		Field corrosion + MCS	Stable, robust optimization	Sensitive to data quality	[66]
	GC-GAN + RF	Generative + Ensemble	Experimental + synthetic data	Solves data scarcity	Needs initial real data	[68]
	GBM	Tree-based ensemble	Lab FCGR test data	Robust to outliers	Class imbalance issues	[76, 111, 152]
	K-means + DT/NN + AHP	Hybrid	ILI data	Multi-criteria integration	Metal loss only	[77]
Service life and degradation prediction	BNN + GIS	BNN	Field corrosion, spatial data	Handles uncertainty	No detailed uncertainty analysis	[84]
	RF + HMOGWO	Physics-informed ensemble	Field corrosion depth	High interpretability	Untested new scenarios	[85]
	BRNN	Recurrent NN	ILI data	Works with incomplete data	Limited field validation	[89]
	Random Survival Forest	Survival ensemble	Field + simulated	Handles censored data	Missing coating/soil data	[91]
Surrogate modelling	GA-BPNN (SPH-FEM)	Optimized NN	SPH-FEM sim	High realism	Narrow parameter range	[98]
	1D-CNN + ANN (ABC-SS)	Hybrid NN	Lab impact data	High accuracy, minimal sensors	Needs field validation	[101]
	TGNN	PINN	Lab + simulated	Works on small datasets	Formula-bias risk	[105]
	MFO-LSSVM + LSTM-DNN	Hybrid optimized SVM+NN	Field + augmented	Robust on small data	Extreme condition gaps	[108]
Defect detection, classification, sizing	TWSC	Signal decomp. + CNN	MFL + eddy current	Noisy data resilience	Narrow defect range	[130]
	D-ResNet + KPCA	PINN	Field env. data	High precision, interpretable	Environment-specific	[154]
	LiDAR + IF + XGBoost	Anomaly detect + Ensemble	LiDAR scans	Cost-efficient	Lab only, single crack	[144]
	VMD + GBDT	Signal decomp. + Ensemble	AE	Strong noise robustness	None stated	[149]
Geohazard-induced integrity	PINN + Transfer Learning	PINN	Physics eqns + FEM/FDM	Meshless nonlinear PGD	Costly training	[139]
	RFE-PSO-AdaBoost + FCM-CRITIC	Hybrid ensemble + clustering	GIS + RS landslides	Objective scoring	Static model	[140]

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decomposition, and classical ML models. Hybrid approaches (e.g., RFE-PSO-AdaBoost + FCM-CRITIC [140], MFO-LSSVM + LSTM-DNN [108], and FVIM-BNN-XGB [127]) combine multiple factors, such as reliability and geohazard. Physics-guided models (PINN + Transfer Learning, TGNN) [105, 139] present accurate results but need large-scale field tests. Signal-decomposition hybrids (TWSC, VMD + GBDT) [130, 149] are effective with noisy data, and interpretable ensembles (GBM, XGBoost) [76, 111, 152] are useful when there are few data samples available. Bayesian GIS-coupled models [84, 87, 140] account for measurement uncertainty, though assessment is not always reliable, with improvements needed in data diversity, generalization, and metric consistency.

Beyond individual accuracy metrics, cross-study comparison reveals that the recurring superior performance of certain architectures arises from their structural compatibility with data modality and uncertainty characteristics. Meta-heuristically optimized neural networks (e.g., LFS-SSA-BPNN [65], LSBES-ELM [66]) outperform conventional feed-forward models in reliability and degradation studies driven by FE simulations or limited field datasets, as global search strategies (Lévy flight, Sparrow Search, Simulated Annealing) mitigate local minima and stabilize learning in highly nonlinear response surfaces. Kernel-based models such as MFO-LSSVM [108] demonstrate superior generalization in small-sample corrosion datasets by balancing bias-variance trade-offs through smooth kernel mappings, while hybridization with LSTM or DNN components enables temporal degradation trends to be captured [108, 116]. In contrast, tree-based ensembles (RF, GBM, XGBoost) [76, 111, 144, 152] excel in heterogeneous inspection and monitoring data owing to their robustness to noise, missing values, and class imbalance, although their extrapolation capability remains limited. Physics-informed and graph-based models (TGNN [105], PINN [139]) show particularly strong performance when governing equations or spatial dependencies constrain learning, explaining their advantage in data-scarce but physics-rich contexts. These trends indicate that reported high R^2 values reflect not only algorithmic sophistication, but alignment between model architecture, data structure, and embedded physical knowledge.

Figure 8 provides a cross-comparison of the best-performing ML approaches related to pipeline management, risk assessment, and integrity monitoring, summarizing the current state of the art in this field. Overall, high performance is observed in demonstrated cases (all models perform well and result in R^2 values over 0.95). Some models, such as RF+HMOGWO [85] for predicting pitting corrosion depth, achieve nearly perfect prediction capabilities ($R^2 = 0.999$), while retaining convenient interpretability. The small difference among the top performers implies that the sector has reached a level of maturity with respect to prediction, such that various architectures (e.g., physics-guided graph NNs (TGNN, $R^2 = 0.9945$) [105] to hybrid signal decomposition plus gradient boosting (Variational Mode Decomposition (VMD) + GBDT, accuracy = 99.44%) [149] can satisfy stringent precision requirements. Incremental improvements are achieved in accuracy; however, further progress may not be feasible due to a lack of comprehensive datasets and validation at full scale, which may hinder generalization beyond certain types of defects or operating conditions.

Domain-guided hybridization emerges as a key factor in achieving the best results. Physics-guided burst pressure predictions attained by TGNN's are particularly well-performing on small datasets [105], while Hybrid CNN-Bo-Transformer [114] achieves a 69.74% error reduction compared to the SVR by integrating the DL interpretability tools (DeepSHAP), combined with corrosion rate modelling. Ensemble methods (e.g., K-means+DT+NN+AHP for defect prioritization, and Relative Risk+NB+DT+SVM+GCN for qualitative risk) adopt multiple decision layers to trade off little precision and flexibility, providing accuracy rates of 97.5% [75, 77]. Even so, the excellent accuracy of RF+HMOGWO requires validation against new corrosion test profiles [85]. Models like XGBoost + LIME [109], are cost-effective and interpretable; however, they may be challenged when used for classifying defect types. The clustering of the visuals, therefore, reflects a hierarchy that depends as much on the problem framing and input data engineering as on mathematical algorithms.

Although Figure 8 represented the best ML techniques, comparisons with lower-performing ones can highlight valuable insights for subsequent investigation. At the lower end of the scale lies BPNN employed for seismic reliability ($R^2 = 0.378$) [65] and SVR for multi-hazard risk prioritization ($R^2 = 0.37$) [161]. Even common ensemble learners delivered poor performance when used without or with limited

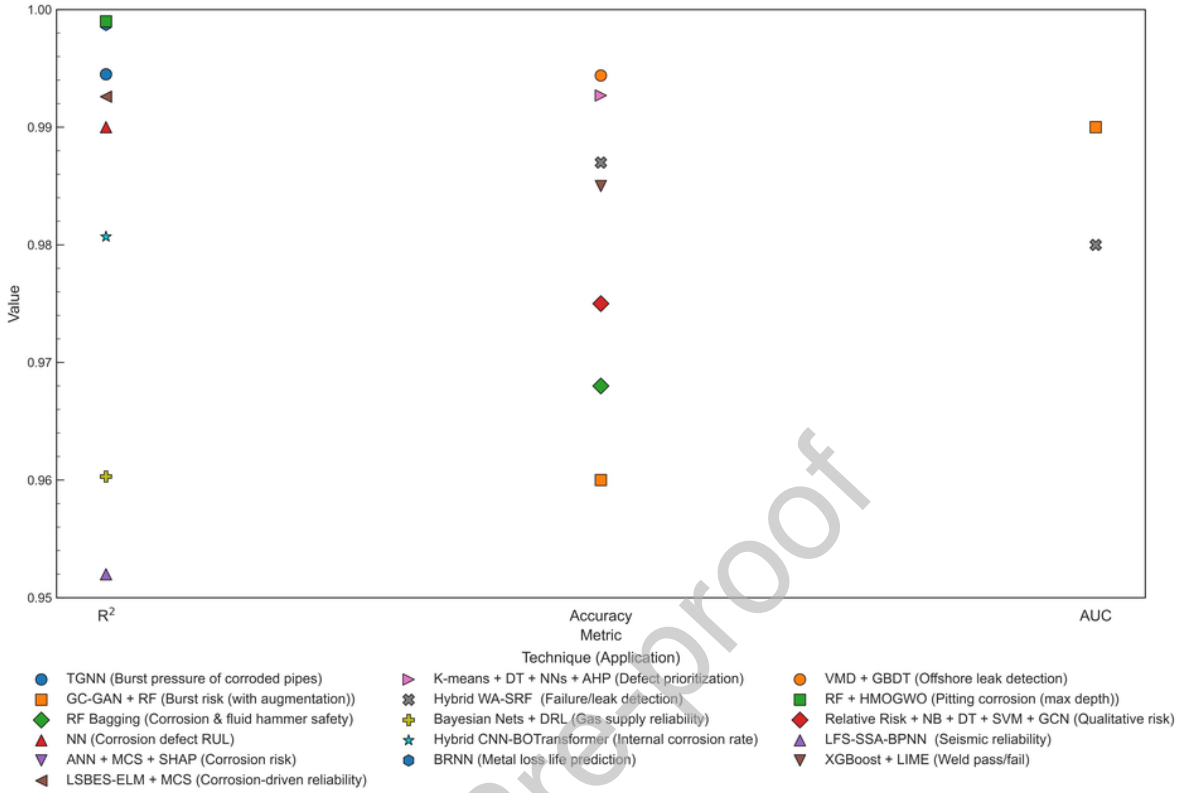


Figure 8. Cross-comparison of high-performance ML techniques for pipeline applications.

tuning, as in RF for corrosion rate estimation (Test $R^2 = 0.56$) [112] and DT for predicting maximum pitting depth ($R^2 = 0.0171$) [120]. The most commonly limited areas were the use of small, homogenized data population, without augmentation or transfer learning, with little integration of domain-specific inputs. In contrast, state-of-the-art ML models tackled these limitations with targeted data enrichment techniques, hybrid learning methods, and physics-informed architectures. This cross-comparison indicates that in future research, the acquisition of large and diverse datasets should be prioritized; physics-guided models, including a rich set of extracted features can be employed; and rigorous validation procedures must be followed to minimize overfitting, strategies that might push a weaker performer into a high-accuracy cluster.

7.2. *Physics-integrated ML taxonomy*

Recent advances in ML-based integrity assessment demonstrate that purely data-driven models are inadequate for pipelines subjected to extreme, transient, and multi-scale loading. The reviewed literature reveals a progression toward physics-integrated ML frameworks, which can be broadly classified into three dominant modes: (i) physics-informed loss enforcement, (ii) theory-guided feature or architecture design, and (iii) hybrid surrogate or residual modeling, with probabilistic reliability embedding increasingly acting as a unifying layer.

Physics integration in ML-based pipeline integrity assessment spans a continuum from explicit constraint enforcement to weakly guided data-driven correction, reflecting different trade-offs between physical fidelity, computational efficiency, and robustness under data scarcity. Strongly constrained approaches, such as PINNs, explicitly embed governing equations and compatibility conditions into the training loss and have shown promising performance for geohazard-induced pipeline response prediction

under permanent ground deformation [138, 139]. However, their current applicability remains largely limited to simplified material behavior and intact geometries. There are challenges in enforcing complex boundary conditions, handling nonlinear plasticity and local buckling, and extending to multi-physics degradation processes. In contrast, theory-guided ML approaches, such as TGNNs [85, 105, 139] and physics-informed feature engineering [117, 119] for burst pressure and corrosion prediction, achieve greater robustness under sparse and noisy data by embedding empirical laws and mechanics-based descriptors directly into model structure or inputs . These models offer improved interpretability and practical stability; nonetheless, their predictive accuracy is inherently tied to the validity and calibration range of the embedded physical assumptions.

A more scalable and operationally mature paradigm emerges in hybrid surrogate and residual-learning frameworks, where ML models either replace computationally expensive physics solvers or systematically correct simplified analytical predictions. Across structural response prediction, corrosion growth, and collapse or burst pressure estimation, ML surrogates trained on FEM datasets enable rapid evaluation with near-physics accuracy [65, 97-109], while residual-based hybrids outperform code-based models such as DNV and ASME by learning model-form discrepancies [88, 103]. ML models, particularly Bayesian networks, embedded within probabilistic reliability frameworks (e.g., MCS) enable tractable uncertainty propagation and decision-oriented integrity assessment under seismic, corrosion, and operational variability [65-67, 69-71, 78, 79]. Nevertheless, most studies emphasize methodological coupling rather than system-level validation, with limited treatment of non-Gaussian extreme loads, interacting degradation mechanisms, and first-passage failure estimation. Together, high-fidelity physics anchors learning models and probabilistic frameworks translate predictions into actionable reliability metrics, positioning uncertainty-aware, validated hybrid physics-ML frameworks as the current state-of-the-art and the most promising pathway for future pipeline integrity management.

7.3. *Key variables in ML techniques*

The input-output parameter relationships summarized in Table 13, along with Figure 9 and Figure 10, clarify key findings regarding pipeline-focused ML models. First, inputs are predominantly multiphysical [108, 113, 156], environmental [66, 85, 111, 114, 116, 140], and geometric parameters [65, 68, 89, 98, 102, 107, 138, 159, 162], while outputs mainly concern corrosion characteristics, stress/strain response, and reliability indices. Second, the heatmap in Figure 9 shows that environmental and operational variables (e.g., pH, chloride, pressure, temperature) are frequently used to predict corrosion rates and reliability indices, with greater emphasis on degradation prediction than on structural capacity. Third, Figure 10 reveals that most ML techniques, especially hybrid and ensemble models, are optimized for predicting mechanical response and defect assessment (e.g., limit state predictions, classification, failure probability). Fourth, data heterogeneity (temporal, spatial, and physical) remains weakly coupled to mechanistic variables, limiting transferability across different pipeline systems. Fifth, though hybrid transformations (e.g., CNN-BO-Transformer, LFS-SSA-BPNN + MCS) bridge feature and mechanistic domains, a gap exists in combining chemical and mechanical variables for simultaneous estimation of degradation, failure, and reliability. Future research should prioritize physics-informed hybridization, integrating stress-strain states, electrochemical factors, and operational parameters into ML algorithms to enable interpretable, probabilistic, and multi-objective integrity prediction, rather than isolated corrosion, stress prediction, or leak-rate estimation.

7.4. *Data origin*

To contextualize the consensus gaps identified in Section 1, the data provenance of the original research studies compiled in Table 2-Table 11 was analyzed, and the origin of data, if explicitly noted in the literature, has been recorded (64 references). Less than half of studies (42.2%) employed field-verified datasets [66, 69, 73, 74, 84, 85, 87, 89, 92-95, 98, 109, 112, 115, 116, 118-122, 134, 145, 152-154, 160, 163], while approximately 5.1 % relied solely on laboratory experiments [101, 144, 150, 151, 158] and 22 % were derived entirely from numerical or simulated data [65, 67, 68, 90, 96, 98, 100, 102-104, 138, 139, 143, 159]. The remaining

Table 13. Input-output mappings of key ML techniques in pipeline engineering.

Model type	Input variables	Predicted output
	Hybrid multi-stage and optimized frameworks	
LFS-SSA-BPNN + FE/MCS [65]	Diameter, wall thickness, pressure, yield and tensile strength	Peak stress under seismic loading; reliability evolution; failure probability
LSBES-ELM + MCS [66]	Redox potential, pH, pipe-to-soil potential, resistivity, density, chloride, moisture, bicarbonate, sulfate, exposure time, coating type, max corrosion depth	Corrosion rate; reliability evolution; failure index; optimal inspection interval
GC-GAN + RF [68]	Diameter, wall thickness, pit depth, pit length, ultimate tensile strength	Burst-failure risk classification (low/high)
SSA-LightGBM [111]	Soil resistivity, redox potential, chloride, sulfate, moisture, pH, salinity, stray current, breakage density, potential, sulfide	Average corrosion rate
MFO-LSSVM + LSTM-DNN [108]	Temperature, pressure, flow rate, temporal corrosion data (time-series)	Remaining service life
LAGAN-GWO-SVR [113]	Flow pattern, flow velocity, inclination angle, CO_2 and H_2S partial pressure, total pressure, gas density, temperature, pH	Internal corrosion rate
RFE-PSO-AdaBoost + FCM-CRITIC [140]	Curvature, elevation, slope, thickness, diameter, coating, burial depth, slope angle.	Landslide susceptibility; pipeline vulnerability
Hybrid WA-SRF [156]	Real-time pressure, flow, temperature, material, failure type/location, inlet-outlet pressure, capacity	Failure and leakage probability; pressure anomaly detection
WOA-XGBoost [159]	Diameter, wall thickness, dent depth, crack depth/length, H_2 blending ratio	Failure pressure of H_2-blended crack-in-dent pipelines
	DL architectures	
CNN-BO-Transformer [114]	Temperature, Cl^- concentration, pressure, CO_2 partial pressure, liquid flow rate, liquid wall shear, gas wall shear, liquid holdup, gas flow rate	Internal corrosion rate
BRNN + Sub-nets [89]	Defect geometry: length, width, depth; nominal wall thickness, pressure, feature ID, wall thinning, corrosion location.	Estimated repair factor; defect dimension classification
ResNet18 + GA-SENet [148]	Time-frequency images (224 × 224 px) derived from denoised eddy-current NDT signals.	Classification of coating-layer defects (e.g., circular pits, scratch grooves)
CCTV + Swin Transformer [145]	High-resolution internal images from CCTV	Localization and classification of internal defects; bounding box, severity, and repair indication
VMD-BiLSTM (MSDBO) [117]	Eight features from TEM detection: time-domain (max, min, mean, slope) and frequency-domain (min, max, mean, median)	Corrosion depth in buried pipelines
Hybrid Transformer-LSTM [116]	Moisture, bicarbonate, chloride, sulfate ion content, pH, soil resistivity	External corrosion rate and depth
	Ensemble tree-based approaches	
RF (Augmented) [112]	Statistical descriptors of pipe-to-soil off-potential	Buried steel corrosion rate
XGBoost+LIME [109]	Potential accident cause features	Ten-year survival curves; girth-weld failure probability
GBRT (FE surrogate) [102]	Pipe geometry (D, t); defect geometry (f, w, h); hole parameters (d, L, C)	Collapse strength of screen pipes
Extra Trees + SMOTE [107]	Material grade, diameter, defect depth, defect length	Burst pressure
RF + CatBoost (Inspection) [162]	Weld type and geometry	Weld inspection ranking
	Physics-informed models	
GA-BP NN (RBF-SVR hybrid) [98]	Pipe diameter, pressure, burial depth, landslide volume, landslide width	Von Mises stress at critical girth-weld section
LSBES-ELM-MCS [97]	Landslide displacement, width, cohesion, friction angle, wall thickness, internal pressure	Maximum von Mises stress; probability of failure; reliability index
HMOGWO-RF [85]	pH, pipe-soil potential, bulk density, bicarbonate, redox potential to pH ratio	Maximum pitting corrosion depth

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PINN + Transfer learning [138]	Pipe location, PGD magnitude and angle, geometry, material modulus, soil-pipe interaction	Axial and lateral displacement; strain fields; strain demand distribution
	Signal decomposition + ML Hybrids	
CWT + Optimized DBN-GA-LSSVM [150]	Enhanced leak-induced scalograms from CWT acoustic-emission signals, denoised via NLM and AHE	Leak detection: binary classification (leak / non-leak)
AE (VMD + GBDT) [149]	Ten time-domain and ten frequency-domain features from four intrinsic-mode functions of AE data	Leak-event identification; noise-versus-leak discrimination
TWSC (STFT + CWT CNN) [130]	Time-frequency fusion features combining STFT and CWT of magnetic-excitation defect signals	Defect-type classification (pitting, straight crack, oblique crack, etc.)
	Classical ML and shallow learners	
RBF kernel [132]	Wave area, peak-to-valley amplitude, valley-to-valley spacing (axial and radial components)	Defect classification/sizing
KNN/DT [75]	Corrosion, leakage, materials, atmosphere, surface, earth movements, minimum depth cover, activity level, coating, product corrosivity, operating pressure (for dispersion), population density, and environmental considerations	Risk level (high, medium, or low risk classification)
SVR [161]	Pipeline attributes, operating pressure, maintenance history, leakage/diffusion/fire/explosion data	Failure probability; accident consequence; risk level; future risk state.
	Bayesian and uncertainty-focused models	
BN + BPNN (Noisy-OR) [94]	Prior probabilities from PHMSA failure records: corrosion, external influence, weld, joint, equipment defect	Failure probability
Bayesian Inference + DRL Fusion [146]	Eddy-current amplitude signals affected by speed, lift-off, angle, defect size	Anomaly localization and classification (defect, weld, bend)

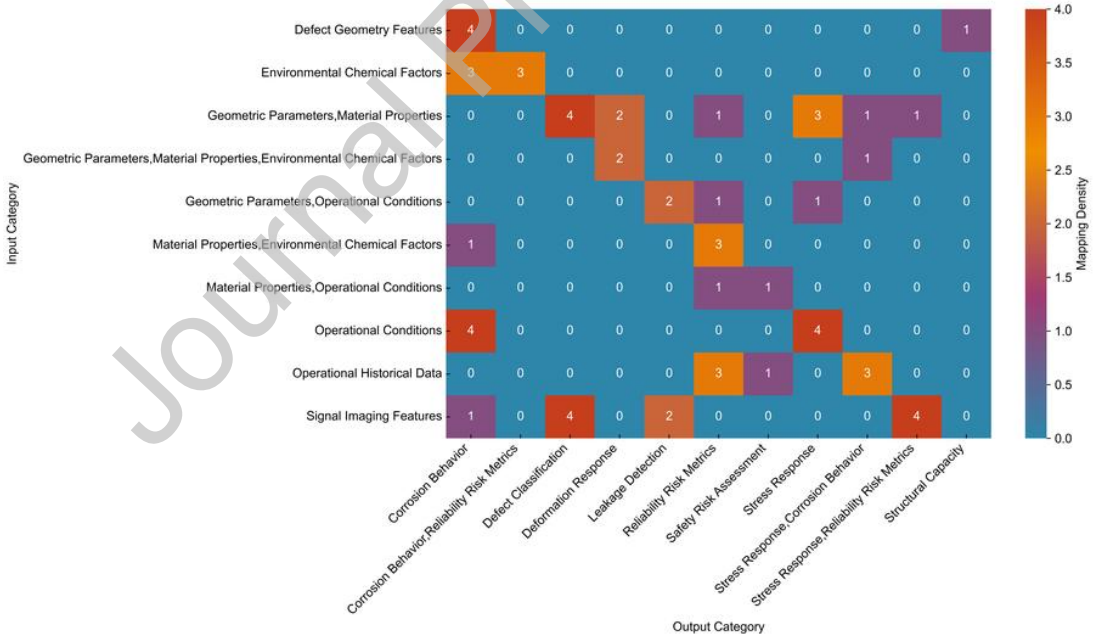


Figure 9. Matrix heatmap of input-output mapping density.

28.1% used hybrid or augmented datasets, integrating limited field or laboratory information with synthetic or FE-generated data [88, 91, 106, 108, 111, 113, 114, 117, 123, 129, 130, 132, 142, 146, 148, 149, 157]. This distribution substantiates the identified gap in field validation and demonstrates that most high-accuracy ML models

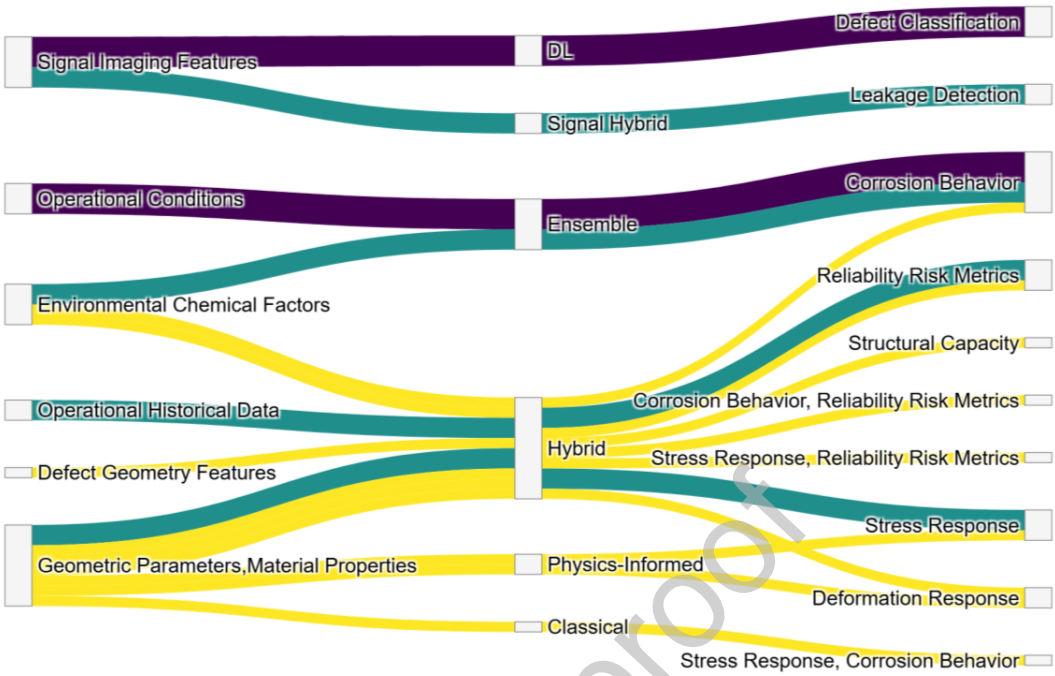


Figure 10. Directional mapping of representative model pathways across input-output domains

remained grounded in constrained or synthetic environments, limiting their generalization to operational pipeline systems. This lack of sufficient validation realism provides a key explanatory link between strong benchmark performance and the persistent challenges of real-world deployment.

7.5. *Data-volume distribution and scalability*

The heatmaps shown in Figure 11 provide a detailed breakdown of the data range used for different applications and ML families. The majority of studies are grouped in small (<1k) and medium (1k-10k) dataset bins, with 100-1k accounting for a total of 35 studies. The prediction of corrosion rate and maximum depth accounts for the dominant application, with the most widely used techniques being tree-based models (30 studies), reflecting their popular performance on low-volume structured data sources. DL architectures are more frequent in mid-to-large dataset bins, whereas PIML and reinforcement learning models occur rarely (with distributed representation), limiting their current use to large-scale predictive tasks. This imbalance supports earlier observations: models trained on small datasets, even when highly accurate, risk overfitting and reduced generalization unless supported by augmentation, transfer learning, or enriched domain features. To this end, PIML techniques should be employed when data is insufficient, by which the physics of the problem can be infused into the NN architecture, allowing the residual loss to tune the error and enhance interpretability. Patterns in data distribution indicate three strategic directions: targeted large-volume data acquisition in underperforming application domains to support robust full-scale validation, careful management of model capacity with respect to dataset size to prevent the inefficient use of high-complexity models, and systematic exploration of bin gaps to accelerate the adoption of PINN and reinforcement learning.

7.6. *Frequent research topics and gaps*

Taken together, Figure 12, and Figure 13 provide a detailed, quantitative breakdown of the ML techniques and gaps in pipeline research, revealing focused areas and persistent blind spots. Overall, NNs, tree-based models, and physics-informed hybrid models prevail in the research space with an

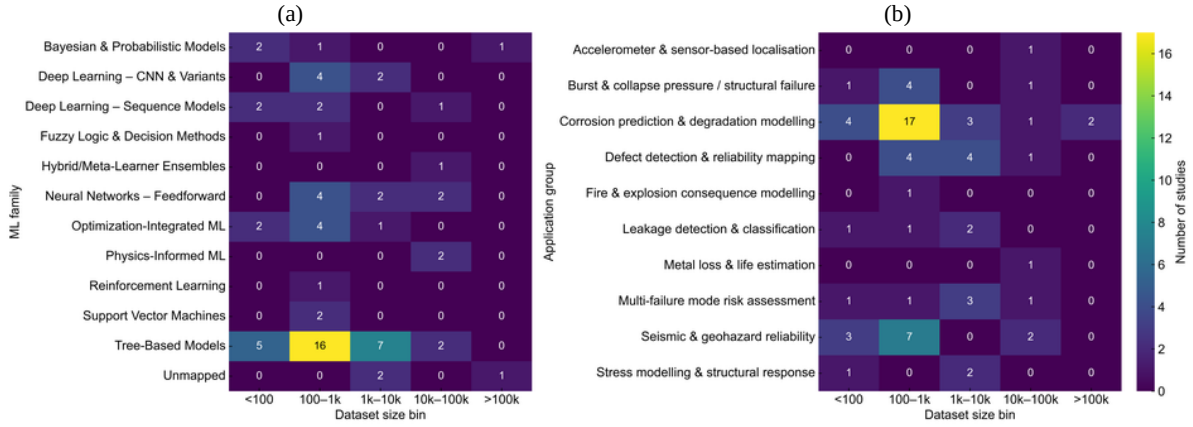


Figure 11. Frequency distribution of dataset size bins: (a) by application group, (b) by ML family.

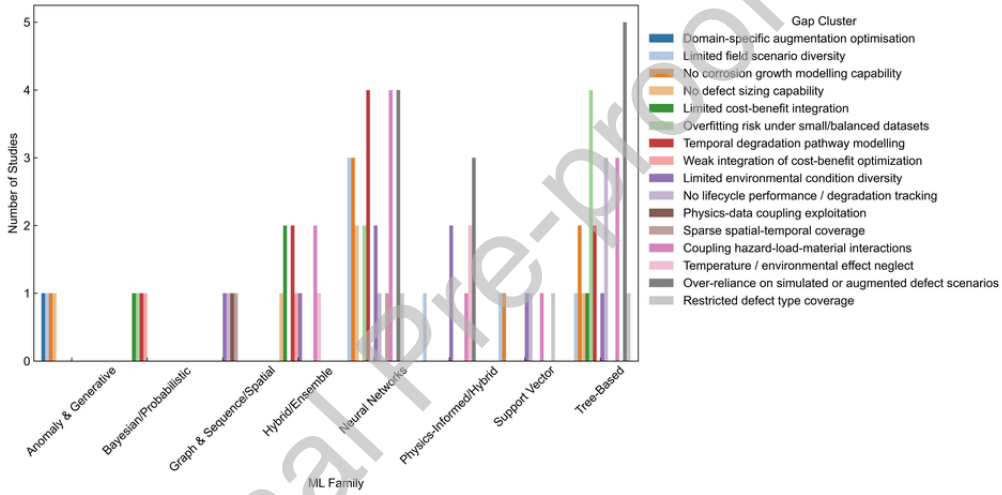


Figure 12. Distribution of research gaps across ML technique families in pipeline integrity, highlighting gap diversity and frequency.

overdependence on synthetic, limited datasets and the risk of model bias. This imbalance is illustrated by the gap distribution shown in Figure 12, where commonly used families tend to exhibit high-frequency limitations, including a lack of sufficient field data, temporal degradation pathway modeling, limited environmental diversity, and poor cost-benefit integration, among others. Figure 13 sharpens the critique, highlighting common research topics and associated research gaps, with over-reliance on simulated or augmented defect scenarios, coupling hazard-load-material interaction, and limited CBA among the most connected and therefore most impactful gaps, which span nearly all application domains.

The summary of gaps and future research directions (Figure 14) highlights the identified key concerns, including hazard-specific overfitting in ML techniques coupled with reliability frameworks, the weak coupling of economic metrics with uncertainty quantifications, synthetic defect scenarios with field validation limited to a few degradation processes and defective structures, and static/single hazard degradation modelling, as well as black-box architectures and insufficient multi-sensor data fusion. These challenges, frequently repeated across several ML families, have been intensified by data scarcity, imbalance, and the absence of defect-interaction modelling, weakening generalizability, trust, and applicability in diverse operational contexts. Several measures can be taken to address these issues. Hazard-inclusive, physics-informed predictive architectures should be developed, with probabilistic ML

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Figure 13. Frequent topics and critical research gaps ranked by connection frequency.

integrated into robust cost-benefit and uncertainty analyses. Multi-source, experimentally validated datasets require expansion, and dynamic hazard-degradation pathway modeling should be studied. More sophisticated constitutive modelling of pipeline infrastructure and environment (elastoplasticicty, anisotropy, time-dependent loading, etc.) would enable more accurate and generalizable predictions of PINN integrated into partial differential equations. Overall, the next generation of AI-assisted pipeline integrity tools calls for coupling methodological diversification with interpretability, transfer learning for cross-domain adaptation, rigorous field validation, and sustainable data acquisition, so that they successfully deliver actionable, transparent, and resilient decision support across the full spectrum of geohazard and risk scenarios.

7.7. *Roadmap and recommendations*

Future progress in ML-assisted pipeline integrity assessment must be evaluated not by algorithmic accuracy alone, but by demonstrable decision readiness under regulatory and operational constraints. Existing studies remain dominated by synthetic or narrowly scoped datasets, static hazard assumptions, and performance metrics weakly connected to integrity decision criteria, limiting transferability and trust in real-world deployment. A structured roadmap is therefore required to align methodological innovation with reliability-based integrity management, ensuring that ML outputs can be audited, calibrated, and interpreted

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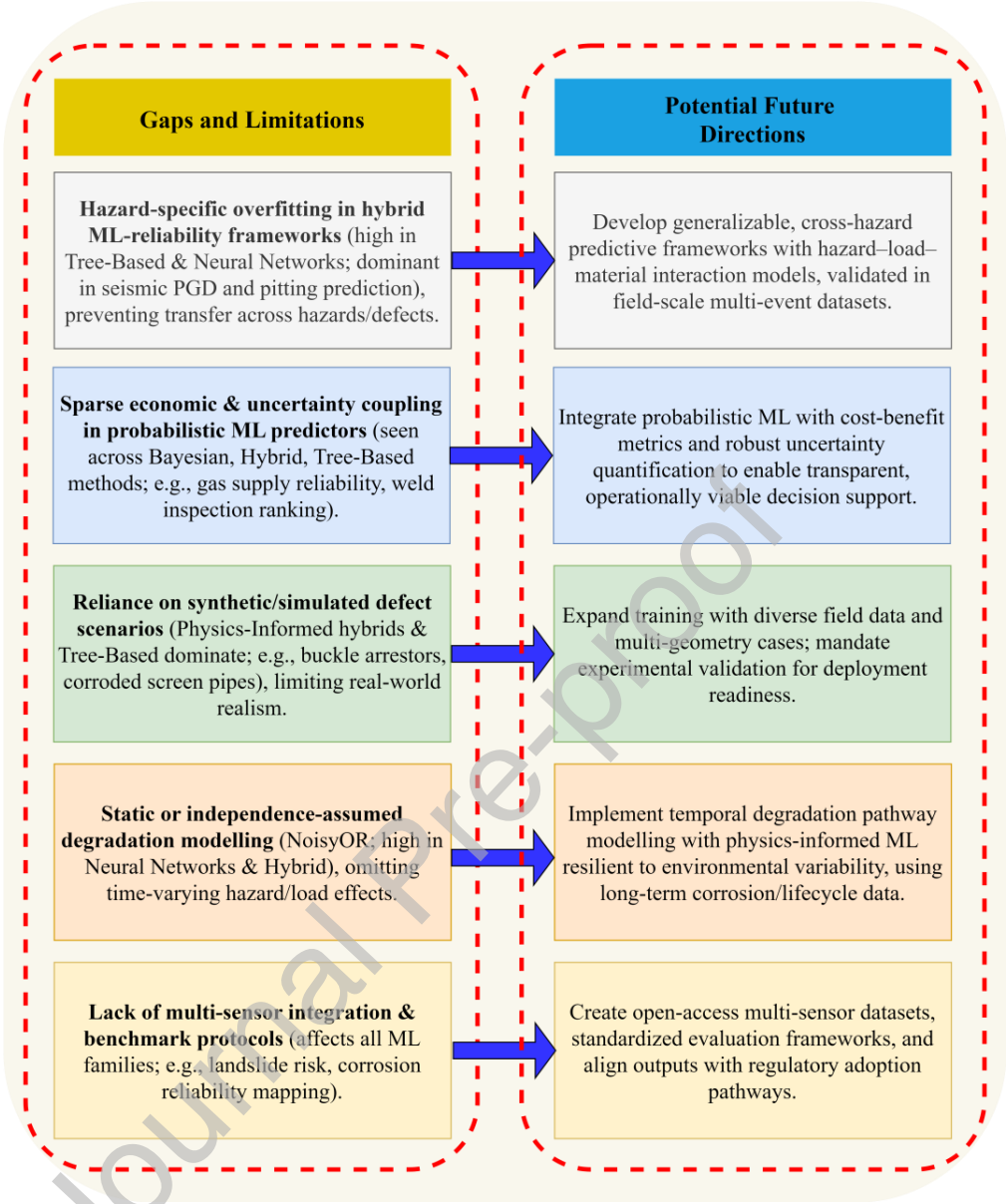


Figure 14. Summary of key research gaps and future research directions in ML-assisted pipeline integrity management.

within established pipeline risk and safety frameworks. Therefore, a decision-matrix roadmap for ML adoption in pipeline integrity has been provided in Table 14.

The three research frontiers include large-scale benchmark datasets, physics-informed and interpretable ML frameworks, and standardized evaluation and field-level validation. For researchers, priority lies in expanding multi-source datasets capturing coupled corrosion, fatigue, geohazard, and impact mechanisms, with explicit uncertainty propagation and probabilistic failure labeling [43, 168]. For operators, emphasis shifts toward robustness under sparse sensing, stability under temporal updating, and integration with digital-twin and reliability workflows for inspection and maintenance planning [39, 169]. For regulators, acceptance depends on evidence that ML-assisted predictions preserve code-level conservatism, reliability targets, and safety margins across representative operating envelopes [170].

Table 14. Decision-matrix roadmap for ML adoption in pipeline integrity.

Research frontier	Researchers	Operators	Regulators
Large-scale benchmark datasets	Multi-hazard, multi-defect datasets with field-verified failure labels and time-dependent degradation	Demonstrated robustness to missing, noisy, and drifting ILI and sensor data	Traceability of datasets to representative operating and environmental envelopes
Physics-informed & interpretable ML	Explicit coupling with governing equations, degradation laws, or limit states; sensitivity analysis	Stable predictions under extrapolation; interpretable drivers of alarms	Evidence that physics guidance reduces unsafe extrapolation
Standardized evaluation protocols	Reliability-aware validation and uncertainty propagation	Calibration of ML outputs to inspection and repair thresholds	Compliance with code-compatible reliability and safety margins

Among these frontiers, standardized evaluation protocols constitute the principal bottleneck for regulatory uptake. Validation must therefore extend beyond R^2, RMSE or classification accuracy toward reliability-consistent metrics that quantify uncertainty, failure likelihood, and decision risk under extreme and low-probability scenarios [171]. Without such alignment, even physics-informed or hybrid models risk remaining analytical surrogates rather than certifiable decision-support tools.

ML models are most effectively adopted when they predict intermediate engineering variables already embedded in standards, such as burst pressure, stress intensity factors, corrosion growth rate, or failure pressure ratios. These outputs can then be substituted into limit-state formulations based on standard codes without altering the decision logic [172, 173]. ML and ML-FEA hybrids reduce the excessive conservatism of semi-empirical codes, particularly for interacting corrosion defects and combined loading, but this accuracy gain is only actionable when translated into safety factors (e.g., pressure ratios, Estimated Repair Factor) or reliability targets already used to trigger repair and inspection decisions [174]. Probabilistic ML and Bayesian frameworks further enable this translation by mapping predictive uncertainty into probability-of-failure or reliability index measures, which align naturally with risk-based inspection intervals and maintenance planning prescribed in API RP 579 and ASME B31.8S workflows [175, 176]. However, ML models reporting only RMSE-optimized strength predictions remain incompatible with regulatory practice, regardless of accuracy [177]. Consequently, ML should be positioned not as a replacement for standards, but as a calibrated surrogate layer that updates code inputs, safety margins, and inspection intervals in a reliability-consistent framework, preserving traceability, auditability, and regulatory acceptance [178, 179].

8. Concluding remarks

ML for PIM has progressed from empirical curve-fitting to hybrid [65, 66, 68, 108, 111, 113, 140, 156, 159] and physics-informed models [85, 97, 98], yet the progress has been limited by fragmented data ecosystems and limited mechanistic coupling. This review reveals that most ML developments in pipeline engineering have concentrated on corrosion, leakage, and structural reliability prediction, while multi-hazard, interacting defects, and lifecycle integrity analyses remain comparatively underexplored. Pipeline integrity assessment has been comprehensively studied using high-performance metaheuristics and surrogate models; nonetheless, reliability-based design and maintenance/inspection planning have received moderate attention. There is a research trend toward interpretable hybridization techniques that account for parameter uncertainty using probabilistic algorithms [65-72, 94, 146], coupling DL, metaheuristics, and physical laws to overcome the black-box barrier that has hindered industrial trust and adoption.

Future studies should integrate electrochemical, geometric, and mechanical parameters within multi-objective frameworks, enabling reliability prediction of defective pipes containing combined defects (e.g., corrosion, crack, dent, etc.) as a continuous lifecycle process rather than isolated event estimation. The foundational issue of data fragmentation can be addressed by acquiring large, heterogeneous, and validated datasets using the latest techniques, such as LiDAR [144], CCTV [145], electromagnetic techniques [146, 148], AE [149, 150], GIS [84, 87, 140, 160, 161], numerical methods (FEM [98, 102, 103, 138, 139, 142, 143, 159, 180], CFD [96, 181], SPH [98], etc.) and synthetic data augmentation [68, 111, 129], accompanied by benchmark-driven evaluation protocols for reproducibility and transferability. The next generation of ML-PIM research should emphasize probabilistic hybridization, cost-benefit coupling, and

multi-sensor fusion, fostering scalable learning frameworks that dynamically respond to spatio-temporal variations in operating conditions. This transformation from information aggregation to risk-informed resilience will lead to digital twin tools that improve predictive maintenance practice, enabling online condition monitoring for continuous reliability assessment. The resulting hybrid intelligent systems will not only elevate technical precision but also bridge operational, economic, and safety dimensions, establishing a self-adapting, physics-consistent predictive infrastructure capable of supporting transparent decision-making across the full pipeline lifecycle.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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